SO3 Converter Ver 1

Converter section ($SO_2 \rightarrow SO_3$) of a sulfur-burning sulfuric-acid plant, 125 t/h H_2SO_4

HAZOP Study Report

Hazard and Operability Study to IEC 61882 — teaching study on the operator training simulator

Client	Department of Chemical Engineering, Chulalongkorn University — teaching study
Document no.	SO3-V1-SA-HZP-001
Revision	A
Study basis	HAZOP tab of the OTS page; screen P&ID SO3-V1-PID-001 rev A; EPC P&ID SO3-PID-001 / SO3-PID-002 rev A

Revision record

Rev	Date	Description	Prepared	Checked	Approved
A	19 Oct 2026	Issued for teaching: HAZOP report of the study on the OTS HAZOP tab			

This is a teaching document. The study uses an example risk matrix and the simulator as its plant; names, signatures and dates are left blank for the class or the reviewing team to complete. It is not a design document for any real plant.

HAZOP close-out summary

43 actions raised from 94 recorded deviations in 10 nodes. The **Action taken** column and the status are completed by the action parties on the close-out sheets of Appendix 3.

Node	Action no.	Raised from	Recommendation	Action by	Action taken	Status	Remark
N1	**R01**	N1.1, N1.7	Auto-start of the spare sulfur pump P-101B on a stop of P-101A or low FI-103, completing before the furnace and bed 1 cool; test the spare monthly.	Instrument / Operations		OPEN	Highest risk: Medium 6
N1/N9	**R02**	N1.2, N1.4, N1.5, N9.4	Sulfur temperature: indication with low and high alarms on the sulfur line at the pump discharge and at the burner (sulfur solidifies near 115–120 °C and becomes viscous above about 160 °C, assumed values); steam-trap and jacket survey on the round sheet.	Process / Operations		OPEN	Highest risk: Medium 9
N1	**R03**	N1.2, N1.11	Sulfur filter: ΔP transmitter to the DCS with a high alarm (today the local gauge PDG-103 only); duplex filters with a preheated spare.	Instrument / Mechanical		OPEN	Highest risk: Medium 6
N1/N2	**R04**	N1.3, N2.4	Cross-limiting of sulfur and combustion air: sulfur demand limited by the measured air (air leads on increase, sulfur leads on decrease), in addition to the FFC-101 ratio station.	Instrument		OPEN	Highest risk: Medium 9
N1	**R05**	N1.3	Verify the sulfur flow meter FI-103 (mass meter, proving, comparison with the pit inventory); the FFC-101 ratio and the production figure both depend on it.	Instrument		OPEN	Highest risk: Medium 9
N1	**R06**	N1.6, N1.7, N1.8	Sulfur pump and line protection: minimum-flow recycle to the pit; confirm discharge check valves; stop the running pump when XV-103 has been closed for a set time; close XV-103 when no sulfur pump runs.	Process / Instrument		OPEN	Highest risk: Medium 9
N1	**R07**	N1.9, N1.10	Written change-over procedure for sulfur filters and pumps (preheated spare, drain, permit); PPE and steam lances at the sulfur area.	Operations		OPEN	Highest risk: Medium 9
N2	**R08**	N2.1	Diversity for I-1: add the K-101 motor-running contact as a second input (1oo2 with FSLL-100) and give FSLL-100 its own flow element, separate from the FI-100 indication.	Instrument		OPEN	Highest risk: High 12
N2	**R09**	N2.2, N2.8	Check the capacity of K-101 and of FV-101/FV-102 at the hottest ambient and at end-of-run bed pressure drops.	Process		OPEN	Highest risk: Medium 8
N2/N4	**R10**	N2.4, N2.5, N3.6, N4.6, N4.7	Low-low O_2 alarm below the AI-105 low alarm with a written response; evaluate a low-O_2 action on the sulfur by LOPA.	Process / Instrument		OPEN	Highest risk: Medium 9
N2	**R11**	N2.6	Engineering limits on the AIC-104 set point and on the FIC-102 remote set point; the SO_2 strength is changed only by written instruction.	Instrument / Operations		OPEN	Highest risk: Medium 6
N2	**R12**	N2.3, N2.7, N2.10	K-101 protection: anti-surge or minimum-flow protection, PI-101 high alarm, motor current alarm, and a non-return damper on the discharge that closes when the blower stops.	Mechanical / Electrical		OPEN	Highest risk: Medium 8
N2	**R13**	N2.9	Moisture analyser on the dried air at the battery limit with a high alarm.	Process		OPEN	Highest risk: Medium 9
N3	**R14**	N3.1, N3.2, N3.7, N3.9	Furnace temperature: 2oo3 thermocouples for TI-106 with a high alarm, and a low-temperature action that closes XV-103 when the furnace is below sulfur ignition while sulfur is admitted (evaluate by LOPA).	Process / Instrument		OPEN	Highest risk: High 12
N3	**R15**	N3.3	H-101 burner management to NFPA 85/86: purge credit before every light-off, flame-scanner proof test, fuel shut-off valve leak test.	Instrument		OPEN	Highest risk: High 10
N3/N8	**R16**	N1.9, N3.7, N3.8	Fixed SO_2 gas detectors at the furnace front, the sulfur pumps, B-101, the converter manways and the acid pumps, alarmed in the control room.	Safety		OPEN	Highest risk: Medium 9

Node	Action no.	Raised from	Recommendation	Action by	Action taken	Status	Remark
N4/N9	**R17**	N4.3, N6.6, N9.3, N9.5	Boiler and economizer water side: confirm that the steam-drum low-low level trip (outside the battery limit) closes XV-103 and add it to the C&E matrix; low-low economizer BFW action and steaming protection.	Process / Instrument		OPEN	Highest risk: Medium 9
N4/N5/N6	**R18**	N4.4, N4.9, N5.7, N5.9, N6.9, N7.3	Tube-leak detection and turnaround inspection: steam/BFW balance and conductivity, leak tests of B-101 and E-201 to E-205, inspection of the converter division plates and bed seals.	Mechanical / Process		OPEN	Highest risk: Medium 9
N4	**R19**	N4.6, N4.8	Analyser validation for AI-104 and AI-105: calibration schedule, sample-system checks and comparison with the SO_2 strength calculated from FI-103 and FI-100.	Instrument		OPEN	Highest risk: Medium 9
N5	**R20**	N4.2, N5.1, N5.2	TSHH-203 on dedicated elements (2oo3), separate from the TI-203 indication and alarm, with bad-PV and deviation alarms between the bed thermocouples and a TI-202 high alarm; LOPA for bed-1 over-temperature.	Instrument / Safety		OPEN	Highest risk: High 12
N4/N5/N7	**R21**	N1.11, N4.5, N5.6, N5.8, N7.6, N7.7	Catalyst management: ΔP trend of every bed, PDI-214 to PDI-244 high alarms, screening at turnarounds, activity sampling and a replacement plan.	Process		OPEN	Highest risk: Medium 9
N5/N7	**R22**	N4.1, N4.2, N5.3, N5.4, N5.5, N7.2, N7.4	Diagnostics on the bypass valves TV-201 to TV-251 (position feedback against controller output, partial stroke at turnaround) and loop-performance monitoring.	Instrument		OPEN	Highest risk: Medium 9
N5/N6/N8	**R23**	N5.10, N6.4, N6.6, N6.10, N8.5	Add DCS alarms where today there is only an indication: TI-207 high, TI-305 high, TI-252 high, AI-303 high, FI-301 high.	Instrument		OPEN	Highest risk: Medium 6
N6/N8	**R24**	N6.1, N8.1, N8.2	Acid pump spares: monthly proof test of the HS-301B auto-start, and an auto-start for P-302B on FSLL-302 like HS-301B.	Instrument / Operations		OPEN	Highest risk: High 12
N6	**R25**	N6.1, N6.2	Level transmitters with low alarm on the acid pump tanks TK-301 and TK-302.	Instrument		OPEN	Highest risk: Medium 8
N6	**R26**	N6.3, N6.4	Dilution water to the IPA acid: double block valves, flow indication and an acid-strength control alarm.	Process		OPEN	Highest risk: Medium 9
N6/N9	**R27**	N6.5, N9.6	Acid-cooler cooling-water flow and ΔT monitoring; anodic-protection alarms.	Process		OPEN	Highest risk: Medium 6
N6/N8	**R28**	N6.7, N8.6	Dew-point margin: minimum gas outlet and BFW inlet temperatures for E-203 to E-205, and a gas bypass for E-205 evaluated for low rates.	Process		OPEN	Highest risk: Medium 6
N6/N8	**R29**	N6.8, N8.9, N10.9	Acid leak protection: spray shields on flanges, safety showers and eyewash at the acid pumps, an acid-leak emergency drill on the simulator.	Safety		OPEN	Highest risk: High 12
N7	**R30**	N7.1	Operating envelope: minimum sulfur rate and SO_2 strength at which bed 4 holds its inlet temperature without start-up help, written into the operating procedure.	Process		OPEN	Highest risk: Medium 8
N7/N10	**R31**	N7.3, N10.3	HV-112: closed-position switch alarmed in the DCS, car-seal register, and the closure signed in the start-up checklist.	Operations / Instrument		OPEN	Highest risk: Medium 9
N7/N8	**R32**	N7.8, N8.4	Mist eliminators in T-301 and T-302: ΔP indication and inspection of the candles at every turnaround.	Mechanical		OPEN	Highest risk: Medium 6
N8	**R33**	N2.5, N7.6, N8.3	ASHH-306 and the permit: review the set point, delay and start-up inhibit against the toxic consequence downwind (dispersion study), and write a permit-exceedance response procedure.	Process / Environment		OPEN	Highest risk: High 12

Node	Action no.	Raised from	Recommendation	Action by	Action taken	Status	Remark
N8	**R34**	N8.3, N8.7, N8.8	Stack analysers: fault alarms for AIT-306 and AIT-307, and an independent SO_2 measurement for ASHH-306 (today the trip and the permit alarm share one analyser).	Instrument		OPEN	Highest risk: High 12
N9	**R35**	N9.1	Instrument air: low-pressure alarm, receiver hold-up time, and confirm that XV-103 fails closed on air loss.	Instrument		OPEN	Highest risk: Medium 6
N9	**R36**	N9.2	Restart after a power failure: acid circulation established before the blower; UPS for DCS and SIS.	Electrical / Operations		OPEN	Highest risk: Medium 9
N9	**R37**	N9.7	XV-103: proof test with partial stroke; LOPA for a second, diverse sulfur shut-off (1oo2), because every trip acts on this one valve.	Instrument / Safety		OPEN	Highest risk: Medium 8
N10	**R38**	N10.5, N10.6	Shutdown sweep: end criteria (time and stack SO_2), dry air only, and no prolonged air flow through a cooling converter.	Operations		OPEN	Highest risk: Medium 9
N3/N10	**R39**	N3.2, N3.4, N3.5, N10.1, N10.2, N10.4	Start-up procedure: maximum heat-up rate, H-101 firing limits, sulfur admission at FIC-103 ≤ 25 % in MAN with a ramp, an alarm when H-101 still fires after sulfur admission, and a start-up emission plan.	Operations		OPEN	Highest risk: High 12
N10	**R40**	N6.8, N10.8, N10.9	Emergency procedures: ESD criteria, trip reset only after a first-out review with supervisor authorisation; simulator drills.	Operations		OPEN	Highest risk: High 12
N10	**R41**	N10.10	Isolation and blinding list for the sulfur and gas systems; SO_2 gas test and permit before entry or opening.	Operations / Safety		OPEN	Highest risk: Medium 8
N10	**R42**	N10.1, N10.11	Trip and permissive bypass management: permit, time limit, MOC, bypass status shown on the DCS.	Operations		OPEN	Highest risk: High 12
N4/N5	**R43**	N4.1, N5.1	Review the effects of I-2 TSHH-203: the trip stops the sulfur only, and with TV-201 wide open the bed-1 temperature keeps rising after the trip (hot furnace refractory heats the air, see D3). Consider forcing TV-201 closed and cutting the air on I-2.	Process / Instrument		OPEN	Highest risk: Medium 9

SO3 Converter Ver 1

HAZOP Study Report

Document no. SO3-V1-SA-HZP-001 · Rev A

Department of Chemical Engineering, Chulalongkorn University — teaching study

Approvals

Role	Name	Signature	Date
Prepared by: HAZOP scribe			
Reviewed by: HAZOP leader (chair)			
Checked by: process engineer			
Checked by: instrument / control engineer			
Approved by: plant manager / owner representative			

Names, signatures and dates are intentionally blank: roles only. No person is named in this teaching study.

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Executive summary

The converter section of SO3 Converter Ver 1 (dried air and molten sulfur in; acid, steam and tail gas out) was divided into **10 nodes**: eight process nodes, one utility node and one procedure node. Applying the eleven IEC 61882 guide words gave **94 recorded deviations** (full recording) and **43 recommendations**. Ranked on the example 5 × 5 matrix with the existing safeguards: **0 very high, 8 high, 77 medium and 9 low**.

Key findings

- **N2.1** Flow NO: No air (blower stops) — High 12 (R08)
- **N3.2** Temperature LESS: Furnace below sulfur ignition — High 12 (R14, R39)
- **N5.2** Measurement OTHER THAN: TI-203 element fails low — High 12 (R20)
- **N6.8** Integrity AS WELL AS: Acid leak at the IPA pumps or piping — High 12 (R29, R40)
- **N8.1** Flow NO: No FAT acid circulation — High 12 (R24)
- **N8.3** Composition MORE: Stack SO_2 above the permit — High 12 (R33, R34)
- **N10.11** Safety system AS WELL AS: A trip or permissive left bypassed — High 12 (R42)
- **N3.3** Flame NO: H-101 flame failure during preheat — High 10 (R15)

From the dynamic check on the simulator

- **N2 blower trip (D2):** FSLL-100 trips the sulfur at 00:06 after K-101 stops. The trip works, but I-1 reads the same flow element as the FI-100 indication, so a meter fault defeats both (R08).
- **N1 loss of sulfur (D1):** no trip, because losing sulfur is the safe direction for the SIS. The air keeps sweeping the converter at the ratio floor of 100 kNm^3/h and in 30 min TI-203 falls to 411 °C and TI-233 to 293 °C, below the ≈355 °C strike of the Cs beds. The consequence is operational (a cold restart); a spare-pump auto-start (R01) removes most of it.
- **N4 TV-201 wide open (D3):** TI-203 alarms at 02:05 and TSHH-203 trips at 02:38 (33 s after the alarm). After the trip TI-203 keeps rising to 709 °C: the hot furnace refractory still heats the air and the open bypass sends it to bed 1. I-2 stops the sulfur but nothing closes TV-201 (R43).
- **N5 TI-203 failed low (D5):** the same TV-201 failure now gives no TI-203 alarm and no TSHH-203 trip, because the indication, the alarm and the trip share one element. Bed-1 middle TI-202 reaches 792 °C and the sulfur is only stopped at 26:24 by ASHH-306, after the stack SO_2 has been far above the permit (R20).
- **N2 dilution air lost (D6):** AI-104 rises to 14.0 % SO_2 and AIT-306 to 4121 ppm while TI-203 stays at or below 631 °C: the O_2 shortage limits conversion instead of overheating bed 1. ASHH-306 trips at 07:00, its 5-min delay included (R33).

Actions by party

Action party	Actions (lead or shared)
Instrument	22
Process	15
Operations	12
Safety	5
Mechanical	4
Electrical	2
Environment	1

Recommendations ranked High address bed-1 over-temperature (single shared TI-203 element), loss of air with sulfur flowing, sulfur admitted below ignition, acid leaks and acid loss to the final absorber, stack SO_2 above the permit, trip bypasses and H-101 flame failure; these go to LOPA before the actions are closed.

1 Introduction

1.1 Purpose

This report documents a Hazard and Operability (HAZOP) study of the converter section of SO3 Converter Ver 1, an operator training simulator of the $SO_2 \rightarrow SO_3$ converter of a 1000 kt/y sulfur-burning sulfuric-acid plant (125 t/h H_2SO_4, 8000 h/y). The study is a teaching example: it shows how the method is applied to a real flowsheet, how safeguards are taken from the actual alarm list and trip settings, and how a dynamic model can check the qualitative judgements.

1.2 Unit description

Main air blower K-101 delivers 238.5 kNm^3/h of dried air at 90 °C, split by FV-101 (combustion air, 204.4 kNm^3/h) and FV-102 (dilution air, 34.1 kNm^3/h). Molten sulfur, 40.94 t/h at about 140 °C, is pumped by P-101A/B through a steam-jacketed filter, the SIS shut-off XV-103 and FV-103 to the burner of furnace F-101 (1121 °C, 14 % SO_2). The waste heat boiler B-101 (4.0 MPa g steam, 94 t/h) with hot-gas bypass TV-201 and the dilution air set the bed-1 inlet at 420 °C, 12.0 % SO_2 / 9.0 % O_2. Converter R-201 has five adiabatic beds (V_2O_5–K_2SO_4 beds 1–3, Cs-promoted beds 4–5) in a 3 + 2 double-absorption arrangement with SH-201, E-201 to E-205, the intermediate absorber T-301 and the final absorber T-302. Design conversion is 99.80 %, stack SO_2 292 ppm against a permit of 500 ppm.

1.3 Safety instrumented system

Every trip acts on one final element, XV-103 (sulfur shut-off); the ESD push-button also stops K-101 and H-101. The trips and start permissives below are read from the simulator:

- I-1 FSLL-100 trip (XV-103) at total air < 90 kNm^3/h, 3 s
- I-2 TSHH-203 trip at bed-1 exit > 660 °C, 5 s (same element as TI-203)
- I-3 FSLL-301 trip at IPA acid < 800 m^3/h, 10 s
- I-4 FSLL-302 trip at FAT acid < 500 m^3/h, 10 s
- I-5 ASHH-306 trip at stack SO_2 > 3000 ppm for 5 min (inhibited 60 min after admission)
- I-6 HS ESD push-button (XV-103 closed, K-101 and H-101 stopped)

Start permissives: start permissive TI-106 ≥ 450 °C; start permissive TIC-201 and TI-202 ≥ 400 °C, inlet ≤ 440 °C; start permissive TI-232 ≥ 360 °C; start permissive FI-301 and FI-302 ≥ 80 %; start permissive HV-112 closed; start permissive FIC-103 in MAN ≤ 25 %.

2 Scope

2.1 Boundaries

Included: K-101 and the air split, the molten-sulfur supply from the pump suction to the burner, F-101 with H-101, B-101 gas side with TV-201, the dilution mixing duct, R-201 beds 1–5 with SH-201 and E-201 to E-205, T-301 and T-302 with their acid circulation pumps, the stack and its analysers, the SIS (I-1 to I-6), the utilities as they affect the unit, and the start-up, shutdown, hot-standby, trip-reset and emergency procedures.

Excluded (battery limits): air-drying tower, sulfur pit and melting, acid coolers, product acid and the acid-strength control system, steam drum and steam users, BFW system. Their effect on the converter is recorded where it enters a node (as a cause), not examined in itself.

2.2 Nodes

Node	Name	Design intent	Rows
N1	Molten sulfur supply: P-101A/B → filter → XV-103 → FV-103 → F-101 burner	Deliver 40.94 t/h of molten sulfur at about 140 °C from the sulfur pit (outside the battery limit) with P-101A (P-101B spare) through the steam-jacketed filter, the SIS shut-off XV-103 and the flow valve FV-103 (FIC-103, fail closed) to the burner of F-101.	11
N2	Air supply K-101 and air split FV-101 (combustion) / FV-102 (dilution)	K-101 delivers 238.5 kNm^3/h of dried air at 90 °C (PI-101 60 kPa g, ≈4.2 MW). FV-101 (fail open, FIC-101 in CAS from the FFC-101 ratio 4.994 kNm^3/t) sends 204.4 kNm^3/h to the furnace; FV-102 (fail open, FIC-102 in CAS from AIC-104) sends 34.1 kNm^3/h of dilution air to the bed-1 inlet.	10
N3	Sulfur furnace F-101 with preheat burner H-101	Burn 40.94 t/h of sulfur in 204.4 kNm^3/h of air to SO_2 gas at 1121 °C, 14 % SO_2, 7 % O_2 (105.3 MW). H-101 (0–30 MW, HIC-109) heats the furnace and the converter during start-up only.	9
N4	Waste heat boiler B-101, hot-gas bypass TV-201 and dilution to bed 1	Cool the furnace gas from 1121 to 274 °C in fire-tube boiler B-101 (4.0 MPa g steam, 94 t/h, ≈45 MW). TV-201 (fail closed, TIC-201) bypasses hot gas, and the dilution air sets the bed-1 inlet at 420 °C, 12.0 % SO_2 / 9.0 % O_2, 238.5 kNm^3/h, PI-200 35 kPa g.	9
N5	Converter first pass: beds 1–3 with SH-201, E-201 and E-202 hot sides	Convert SO_2 to 95.83 % over beds 1–3 (V_2O_5–K_2SO_4, 33.6 / 37.4 / 45.1 t) with interpass cooling: bed 1 420 → 631 °C; SH-201 (gas 631 → 598 °C, steam to 292 °C, bypass TV-231 / TIC-231); E-201 hot side (bypass TV-211, TIC-211 bed-2 inlet 445 °C); bed 2 445 → 528 °C; E-202 hot side (bypass TV-221, TIC-221 bed-3 inlet 440 °C); bed 3 440 → 465 °C.	10
N6	Economizer E-203, intermediate absorber T-301 and IPA acid circulation	Cool the bed-3 gas to 190 °C in economizer E-203 (bypass TV-251, TIC-251) and absorb the SO_3 in T-301 with 1600 m^3/h of 98.5 % acid at 80 °C (P-301A, P-301B spare with the HS-301B auto-start on FSLL-301); the gas leaves at 83 °C to the E-202 cold side.	10
N7	Return gas (E-202 / E-201 cold sides), beds 4–5 and E-204	Reheat the absorbed gas from 83 to 204 °C in E-202 and to 425 °C in E-201 (the bed-4 inlet depends on the heat beds 1–2 give up, trimmed by TV-231 at SH-201); convert to 99.70 % in Cs-promoted bed 4 (425 → 441 °C) and 99.80 % in bed 5 (428 → 427 °C) after E-204 (bypass TV-241, TIC-241).	8
N8	Economizer E-205, final absorber T-302, stack ST-301 and stack analysers	Cool the bed-5 gas to 160 °C in E-205 (no bypass) and absorb the remaining SO_3 in T-302 with 1000 m^3/h of acid (P-302A, P-302B manual spare); tail gas to the stack at 292 ppm SO_2 (permit 500 ppm, AIT-306) and 10.5 mg/Nm^3 acid mist (AIT-307); XI-308 99.80 %; FI-309 125 t/h acid.	9
N9	Utilities: instrument air, power, BFW, steam and cooling water	Supply instrument air to every control valve (each goes to its fail position on loss), electric power to K-101 (≈4.2 MW) and the pumps, BFW 150 t/h to E-203 / E-204 / E-205 (FI-253), steam for the sulfur jackets and cooling water for the acid coolers; take 94 t/h of steam from B-101 through SH-201.	7
N10	Procedures: start-up preheat, sulfur admission, shutdown sweep, hot standby, trip reset, ESD	Preheat the furnace and the beds with H-101 (HV-112 open so hot gas also reaches beds 4–5), close HV-112, admit sulfur through the SIS when the start permissives hold, line out; stop sulfur, sweep 5 min at about 100 kNm^3/h and stop K-101 and the acid pumps to hold the beds hot (hot standby); emergency shutdown with the ESD push-button.	11

2.3 Node map

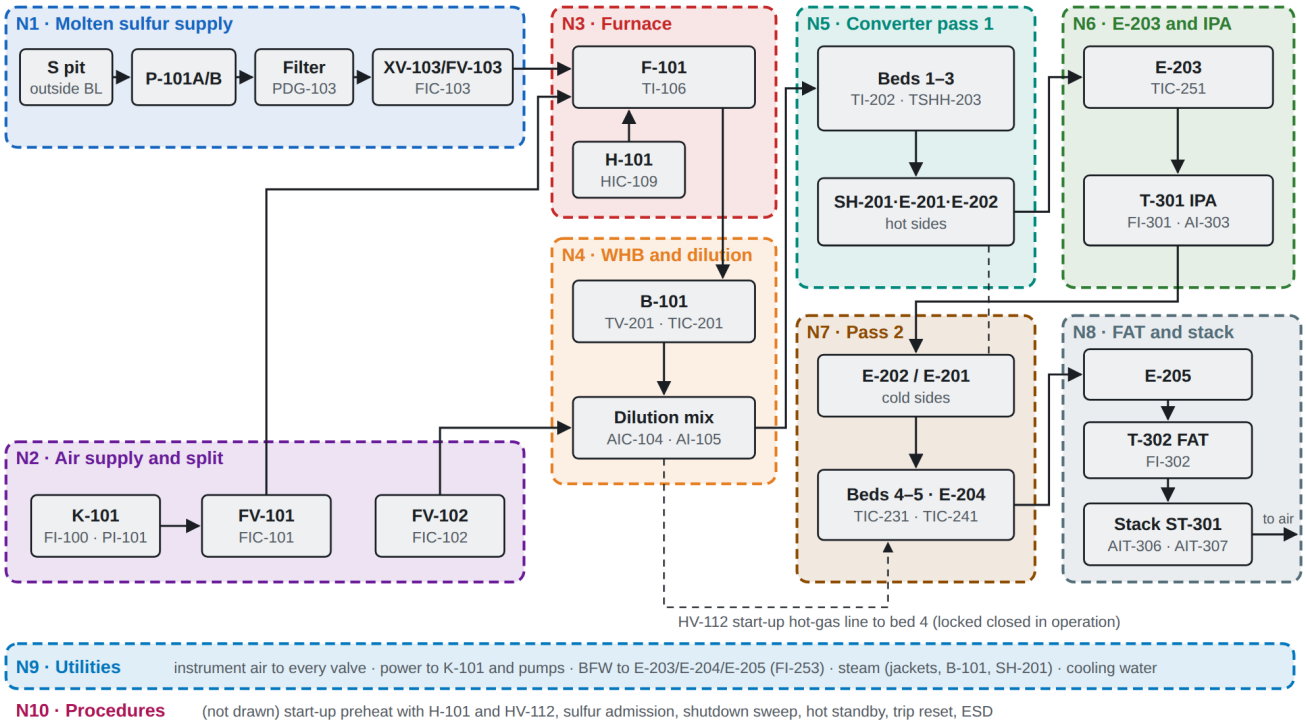


Figure 2.1 — Node map: coloured dashed boxes mark the nodes over simplified equipment blocks; N10 (procedures) is not drawn. A larger copy is in Appendix 2.

3 Methodology

3.1 Standard and phases

The study follows IEC 61882:2016 with the worksheet practice of the CCPS *Guidelines for Hazard Evaluation Procedures*; scenarios relying on instrumented trips feed LOPA and SIL determination to IEC 61511. Four phases were followed:

- **Definition**: scope, objectives, team and risk matrix.
- **Preparation**: P&ID, PFD and stream table, C&E matrix, SRS, procedures, alarm list; nodes and design intents; parameters and guide words.
- **Examination**: node → design intent → parameter → guide word → deviation → causes → consequences (no credit for safeguards) → existing safeguards → risk → recommendation → next guide word. All meaningful deviations are recorded (full recording).
- **Documentation and follow-up**: worksheets (Appendix 1), recommendation register (section 5.3), action close-out sheets (Appendix 3), revalidation after MOC.

3.2 Guide words and parameters

Guide word	Meaning (IEC 61882)	Example in this study
NO / NOT	Complete negation of the intent	Flow no: No sulfur to the burner (N1.1)
MORE	Quantitative increase	Flow more: More sulfur than intended (N1.3)
LESS	Quantitative decrease	Flow less: Less sulfur than set (N1.2)
AS WELL AS	Qualitative increase: something extra present	Integrity as well as: Molten sulfur leak (N1.9)
PART OF	Qualitative decrease: only part of the intent achieved	Composition part of: Incomplete sulfur combustion (N3.6)
REVERSE	Logical opposite of the intent	Flow reverse: Hot furnace gas back into the sulfur line (N1.8)
OTHER THAN	Complete substitution: something else happens	Maintenance other than: Filter or pump change-over done incorrectly (N1.10)
EARLY	Relative to clock time	Trip reset early: SIS reset and sulfur re-admitted before the cause is cleared (N10.8)
LATE	Relative to clock time	Hot standby late: Restart after the beds have cooled (N10.7)
BEFORE	Relative to order or sequence	Sequence before: Sulfur admitted before acid circulates (N10.2)
AFTER	Relative to order or sequence	Sequence after: HV-112 closed late or not at all (N10.3)

Parameters: flow, pressure, temperature (process gas, catalyst bed, sulfur, acid), composition (SO_2, O_2, acid strength, moisture, contaminants), catalyst activity, heat transfer, measurement, control, integrity (containment), utility, safety system, maintenance; for procedures: preheat, sulfur admission, sequence, sweep, hot standby, trip reset, emergency shutdown. Only physically meaningful deviations are recorded.

3.3 Risk matrix (example for teaching)

Risk R = S × L. Severity S ranks the worst credible consequence with no credit for safeguards; likelihood L ranks its frequency with the existing safeguards working. Numbers in brackets are the deviations of this study in each cell.

S \ L	L1	L2	L3	L4	L5
S5	**5**	**10 (1)**	**15**	**20**	**25**
S4	**4**	**8 (9)**	**12 (6)**	**16**	**20**
S3	**3**	**6 (12)**	**9 (21)**	**12 (1)**	**15**
S2	**2**	**4 (4)**	**6 (33)**	**8 (2)**	**10**
S1	**1**	**2**	**3 (3)**	**4 (2)**	**5**

S	Severity (worst credible consequence)
1	First aid; minor upset; no release
2	Medical treatment; outage under a day; small release inside the unit; short permit exceedance
3	Lost-time injury; shutdown for days; partial catalyst damage; reportable SO_2 or acid-mist exceedance
4	Permanent disability or a single fatality; major damage (converter, B-101); large toxic release on site
5	Multiple fatalities; explosion; toxic release beyond the site

L	Likelihood with existing safeguards
1	Remote: not expected in the life of the plant ($< 10^{-4}$ /y)
2	Unlikely: known in industry (10^{-4}–10^{-3} /y)
3	Possible: could happen in the plant life (10^{-3}–10^{-2} /y)
4	Likely: several times in the plant life (10^{-2}–10^{-1} /y)

L	Likelihood with existing safeguards
5	Frequent: yearly or more

Risk	Action
1–4 Low	Acceptable
5–9 Medium	Reduce if practicable (ALARP); review safeguards
10–16 High	Recommendation required; consider LOPA
20–25 Very high	Not acceptable; act before operation

4 Sessions

4.1 Schedule

Session	Nodes examined	Date	Duration	Attendance
1	Kick-off, scope, risk matrix; N1, N2			
2	N3, N4			
3	N5, N6			
4	N7, N8			
5	N9, N10; review of recommendations			

Dates, durations and attendance are left blank for the class or review team to fill in.

4.2 Team

Role	Responsibility	Name	Organisation
HAZOP leader (chair)	Independent of the design; runs the method, keeps the team on the node, decides when a deviation is covered		
Scribe (recorder)	Writes the worksheet live, reads back every recommendation		
Process engineer (designer)	Design intent, heat and material balance, bed temperature and conversion profile, limits		
Operations representative	How the converter is really run: start-up preheat, sulfur admission, hot standby, upsets, rounds		
Instrument / control engineer	Loops, fail positions on instrument-air loss, SIS logic (I-1 to I-6), trip independence		
Safety / risk engineer	Risk matrix, SO_2/SO_3 toxic consequences, LOPA follow-up, regulations and permit		
Specialists as needed	Catalyst vendor, boiler (B-101) and exchanger mechanical, acid-plant materials, environmental		

4.3 Documents reviewed

Document	Used for
P&ID tab (screen P&ID) and EPC P&ID SO3-PID-001 / SO3-PID-002 rev A	Equipment, lines, instruments, valves and fail positions
Process tab and Design tab	Design intent, stream conditions, bed profile, duties, catalyst loads
BFD and PFD tabs	Battery limits, stream table
Control loop design tab	Loops, cascades, ratio station, tuning
Safety system (SIS) tab and Workbook tab	Trips I-1 to I-6, delays, start permissives, alarm list (ALM)
C&E matrix and SRS of SIF tabs	Cause-and-effect of every trip, safety requirements
Exercises tab, SOP tab	Start-up, hot restart, shutdown, hot standby and ESD procedures
HAZID tab	Hazards identified before this study

4.4 Guide-word coverage

Node	NO/NOT	MORE	LESS	AS WELL AS	PART OF	REVERSE	OTHER THAN	EARLY	LATE	BEFORE	AFTER	Rows
N1	1	3	3	2	·	1	1	·	·	·	·	11
N2	1	4	3	1	·	1	·	·	·	·	·	10
N3	1	3	1	2	1	·	1	·	·	·	·	9
N4	1	3	3	1	·	·	1	·	·	·	·	9
N5	·	4	2	1	1	·	2	·	·	·	·	10
N6	1	4	3	2	·	·	·	·	·	·	·	10
N7	·	3	3	2	·	·	·	·	·	·	·	8
N8	1	3	2	1	·	·	2	·	·	·	·	9
N9	5	1	1	·	·	·	·	·	·	·	·	7
N10	·	2	2	1	·	·	1	1	2	1	1	11
All	11	30	23	13	2	2	8	1	2	1	1	94

A dot means the guide word was applied but gave no meaningful deviation. EARLY, LATE, BEFORE and AFTER apply mainly to the procedure node N10.

5 Findings

5.1 Risk profile per node

Node	Name	Rows	Very high	High	Medium	Low	Highest R
N1	Molten sulfur supply	11	0	0	10	1	9
N2	Air supply K-101 and air split FV-101 (combustion) / FV-102 (dilution)	10	0	1	8	1	12
N3	Sulfur furnace F-101 with preheat burner H-101	9	0	2	7	0	12
N4	Waste heat boiler B-101, hot-gas bypass TV-201 and dilution to bed 1	9	0	0	9	0	9
N5	Converter first pass	10	0	1	9	0	12
N6	Economizer E-203, intermediate absorber T-301 and IPA acid circulation	10	0	1	7	2	12
N7	Return gas (E-202 / E-201 cold sides), beds 4–5 and E-204	8	0	0	6	2	9
N8	Economizer E-205, final absorber T-302, stack ST-301 and stack analysers	9	0	2	7	0	12
N9	Utilities	7	0	0	6	1	9
N10	Procedures	11	0	1	8	2	12
All		94	0	8	77	9	

5.2 High-ranked deviations

Ref	Deviation	Consequence (no credit for safeguards)	Existing safeguards	Risk	Rec.
N2.1	Flow NO: No air (blower stops)	Sulfur keeps flowing into a hot furnace without air: sulfur vapour accumulates in F-101, B-101 and the duct and can ignite violently when air returns; no gas through the beds	• I-1 FSLL-100 trip (XV-103) at total air < 90 kNm^3/h, 3 s • XV-103 closes • FI-100 alarm LO 200 kNm^3/h • I-6 HS ESD push-button (XV-103 closed, K-101 and H-101 stopped)	**High 12**	R08
N3.2	Temperature LESS: Furnace below sulfur ignition	Sulfur does not ignite and pools in the furnace; the pool later ignites at once: temperature surge, sulfur vapour carry-over, SO_2 puff, overpressure	• start permissive TI-106 ≥ 450 °C • start permissive FIC-103 in MAN ≤ 25 % • nothing acts once sulfur is in	**High 12**	R14, R39
N5.2	Measurement OTHER THAN: TI-203 element fails low	The TI-203 alarm and the TSHH-203 trip read the same element, so both are blind: a real bed-1 over-temperature goes undetected (see D5: bed-1 middle near 790 °C, no TSHH-203 trip)	• TI-202 bed-1 middle indication (no alarm) • AIT-306 alarm HI 500 ppm late	**High 12**	R20
N6.8	Integrity AS WELL AS: Acid leak at the IPA pumps or piping	98.5 % acid at 80 °C: severe chemical burns; spill	• I-6 HS ESD push-button (XV-103 closed, K-101 and H-101 stopped) • acid pumps stop from the DCS • PPE, safety showers (assumed)	**High 12**	R29, R40
N8.1	Flow NO: No FAT acid circulation	SO_3 goes up the stack as a dense acid-mist plume that can reach people downwind (see D8: AIT-307 far above its alarm before the trip); tower lining damage	• FI-302 alarm LO 750 m^3/h • I-4 FSLL-302 trip at FAT acid < 500 m^3/h, 10 s • AIT-307 alarm HI 40 mg/Nm^3	**High 12**	R24
N8.3	Composition MORE: Stack SO_2 above the permit	Permit exceeded: fines, complaints; at high levels a health risk downwind	• AIT-306 alarm HI 500 ppm • I-5 ASHH-306 trip at stack SO_2 > 3000 ppm for 5 min (inhibited 60 min after admission)	**High 12**	R33, R34
N10.11	Safety system AS WELL AS: A trip or permissive left bypassed	The protection is missing when the demand comes (over-temperature, loss of air, acid loss)	Administrative control only	**High 12**	R42
N3.3	Flame NO: H-101 flame failure during preheat	Unburned fuel enters the hot furnace; ignition of the accumulated fuel: explosion in F-101 or B-101	• H-101 own flame protection (fuel shut off) • H-101 trips itself when total air < 60 kNm^3/h • purge before relight	**High 10**	R15

5.3 Recommendation register

No.	Recommendation	Node	Raised from	Highest risk	Action by
R01	Auto-start of the spare sulfur pump P-101B on a stop of P-101A or low FI-103, completing before the furnace and bed 1 cool; test the spare monthly.	N1	N1.1, N1.7	**Medium 6**	Instrument / Operations
R02	Sulfur temperature: indication with low and high alarms on the sulfur line at the pump discharge and at the burner (sulfur solidifies near 115–120 °C and becomes viscous above about 160 °C, assumed values); steam-trap and jacket survey on the round sheet.	N1/N9	N1.2, N1.4, N1.5, N9.4	**Medium 9**	Process / Operations
R03	Sulfur filter: ΔP transmitter to the DCS with a high alarm (today the local gauge PDG-103 only); duplex filters with a preheated spare.	N1	N1.2, N1.11	**Medium 6**	Instrument / Mechanical
R04	Cross-limiting of sulfur and combustion air: sulfur demand limited by the measured air (air leads on increase, sulfur leads on decrease), in addition to the FFC-101 ratio station.	N1/N2	N1.3, N2.4	**Medium 9**	Instrument
R05	Verify the sulfur flow meter FI-103 (mass meter, proving, comparison with the pit inventory); the FFC-101 ratio and the production figure both depend on it.	N1	N1.3	**Medium 9**	Instrument
R06	Sulfur pump and line protection: minimum-flow recycle to the pit; confirm discharge check valves; stop the running pump when XV-103 has been closed for a set time; close XV-103 when no sulfur pump runs.	N1	N1.6, N1.7, N1.8	**Medium 9**	Process / Instrument
R07	Written change-over procedure for sulfur filters and pumps (preheated spare, drain, permit); PPE and steam lances at the sulfur area.	N1	N1.9, N1.10	**Medium 9**	Operations
R08	Diversity for I-1: add the K-101 motor-running contact as a second input (1oo2 with FSLL-100) and give FSLL-100 its own flow element, separate from the FI-100 indication.	N2	N2.1	**High 12**	Instrument
R09	Check the capacity of K-101 and of FV-101/FV-102 at the hottest ambient and at end-of-run bed pressure drops.	N2	N2.2, N2.8	**Medium 8**	Process
R10	Low-low O_2 alarm below the AI-105 low alarm with a written response; evaluate a low-O_2 action on the sulfur by LOPA.	N2/N4	N2.4, N2.5, N3.6, N4.6, N4.7	**Medium 9**	Process / Instrument
R11	Engineering limits on the AIC-104 set point and on the FIC-102 remote set point; the SO_2 strength is changed only by written instruction.	N2	N2.6	**Medium 6**	Instrument / Operations
R12	K-101 protection: anti-surge or minimum-flow protection, PI-101 high alarm, motor current alarm, and a non-return damper on the discharge that closes when the blower stops.	N2	N2.3, N2.7, N2.10	**Medium 8**	Mechanical / Electrical
R13	Moisture analyser on the dried air at the battery limit with a high alarm.	N2	N2.9	**Medium 9**	Process
R14	Furnace temperature: 2oo3 thermocouples for TI-106 with a high alarm, and a low-temperature action that closes XV-103 when the furnace is below sulfur ignition while sulfur is admitted (evaluate by LOPA).	N3	N3.1, N3.2, N3.7, N3.9	**High 12**	Process / Instrument
R15	H-101 burner management to NFPA 85/86: purge credit before every light-off, flame-scanner proof test, fuel shut-off valve leak test.	N3	N3.3	**High 10**	Instrument
R16	Fixed SO_2 gas detectors at the furnace front, the sulfur pumps, B-101, the converter manways and the acid pumps, alarmed in the control room.	N3/N8	N1.9, N3.7, N3.8	**Medium 9**	Safety
R17	Boiler and economizer water side: confirm that the steam-drum low-low level trip (outside the battery limit) closes XV-103 and add it to the C&E matrix; low-low economizer BFW action and steaming protection.	N4/N9	N4.3, N6.6, N9.3, N9.5	**Medium 9**	Process / Instrument
R18	Tube-leak detection and turnaround inspection: steam/BFW balance and conductivity, leak tests of B-101 and E-201 to E-205, inspection of the converter division plates and bed seals.	N4/N5/N6	N4.4, N4.9, N5.7, N5.9, N6.9, N7.3	**Medium 9**	Mechanical / Process
R19	Analyser validation for AI-104 and AI-105: calibration schedule, sample-system checks and comparison with the SO_2 strength calculated from FI-103 and FI-100.	N4	N4.6, N4.8	**Medium 9**	Instrument
R20	TSHH-203 on dedicated elements (2oo3), separate from the TI-203 indication and alarm, with bad-PV and deviation alarms between the bed thermocouples and a TI-202 high alarm; LOPA for bed-1 over-temperature.	N5	N4.2, N5.1, N5.2	**High 12**	Instrument / Safety
R21	Catalyst management: ΔP trend of every bed, PDI-214 to PDI-244 high alarms, screening at turnarounds, activity sampling and a replacement plan.	N4/N5/N7	N1.11, N4.5, N5.6, N5.8, N7.6, N7.7	**Medium 9**	Process
R22	Diagnostics on the bypass valves TV-201 to TV-251 (position feedback against controller output, partial stroke at turnaround) and loop-performance monitoring.	N5/N7	N4.1, N4.2, N5.3, N5.4, N5.5, N7.2, N7.4	**Medium 9**	Instrument

No.	Recommendation	Node	Raised from	Highest risk	Action by
R23	Add DCS alarms where today there is only an indication: TI-207 high, TI-305 high, TI-252 high, AI-303 high, FI-301 high.	N5/N6/N8	N5.10, N6.4, N6.6, N6.10, N8.5	**Medium 6**	Instrument
R24	Acid pump spares: monthly proof test of the HS-301B auto-start, and an auto-start for P-302B on FSLL-302 like HS-301B.	N6/N8	N6.1, N8.1, N8.2	**High 12**	Instrument / Operations
R25	Level transmitters with low alarm on the acid pump tanks TK-301 and TK-302.	N6	N6.1, N6.2	**Medium 8**	Instrument
R26	Dilution water to the IPA acid: double block valves, flow indication and an acid-strength control alarm.	N6	N6.3, N6.4	**Medium 9**	Process
R27	Acid-cooler cooling-water flow and ΔT monitoring; anodic-protection alarms.	N6/N9	N6.5, N9.6	**Medium 6**	Process
R28	Dew-point margin: minimum gas outlet and BFW inlet temperatures for E-203 to E-205, and a gas bypass for E-205 evaluated for low rates.	N6/N8	N6.7, N8.6	**Medium 6**	Process
R29	Acid leak protection: spray shields on flanges, safety showers and eyewash at the acid pumps, an acid-leak emergency drill on the simulator.	N6/N8	N6.8, N8.9, N10.9	**High 12**	Safety
R30	Operating envelope: minimum sulfur rate and SO_2 strength at which bed 4 holds its inlet temperature without start-up help, written into the operating procedure.	N7	N7.1	**Medium 8**	Process
R31	HV-112: closed-position switch alarmed in the DCS, car-seal register, and the closure signed in the start-up checklist.	N7/N10	N7.3, N10.3	**Medium 9**	Operations / Instrument
R32	Mist eliminators in T-301 and T-302: ΔP indication and inspection of the candles at every turnaround.	N7/N8	N7.8, N8.4	**Medium 6**	Mechanical
R33	ASHH-306 and the permit: review the set point, delay and start-up inhibit against the toxic consequence downwind (dispersion study), and write a permit-exceedance response procedure.	N8	N2.5, N7.6, N8.3	**High 12**	Process / Environment
R34	Stack analysers: fault alarms for AIT-306 and AIT-307, and an independent SO_2 measurement for ASHH-306 (today the trip and the permit alarm share one analyser).	N8	N8.3, N8.7, N8.8	**High 12**	Instrument
R35	Instrument air: low-pressure alarm, receiver hold-up time, and confirm that XV-103 fails closed on air loss.	N9	N9.1	**Medium 6**	Instrument
R36	Restart after a power failure: acid circulation established before the blower; UPS for DCS and SIS.	N9	N9.2	**Medium 9**	Electrical / Operations
R37	XV-103: proof test with partial stroke; LOPA for a second, diverse sulfur shut-off (1oo2), because every trip acts on this one valve.	N9	N9.7	**Medium 8**	Instrument / Safety
R38	Shutdown sweep: end criteria (time and stack SO_2), dry air only, and no prolonged air flow through a cooling converter.	N10	N10.5, N10.6	**Medium 9**	Operations
R39	Start-up procedure: maximum heat-up rate, H-101 firing limits, sulfur admission at FIC-103 ≤ 25 % in MAN with a ramp, an alarm when H-101 still fires after sulfur admission, and a start-up emission plan.	N3/N10	N3.2, N3.4, N3.5, N10.1, N10.2, N10.4	**High 12**	Operations
R40	Emergency procedures: ESD criteria, trip reset only after a first-out review with supervisor authorisation; simulator drills.	N10	N6.8, N10.8, N10.9	**High 12**	Operations
R41	Isolation and blinding list for the sulfur and gas systems; SO_2 gas test and permit before entry or opening.	N10	N10.10	**Medium 8**	Operations / Safety
R42	Trip and permissive bypass management: permit, time limit, MOC, bypass status shown on the DCS.	N10	N10.1, N10.11	**High 12**	Operations
R43	Review the effects of I-2 TSHH-203: the trip stops the sulfur only, and with TV-201 wide open the bed-1 temperature keeps rising after the trip (hot furnace refractory heats the air, see D3). Consider forcing TV-201 closed and cutting the air on I-2.	N4/N5	N4.1, N5.1	**Medium 9**	Process / Instrument

5.4 Dynamic check on the simulator

Ten deviations were run on copies of the converter model from design conditions with the safeguards active (alarms evaluated every second against the alarm list, SIS trips with their delays, controllers in their normal modes, fail positions) and no operator action, for 30 min or until 2 min after a trip.

Run	Node (row)	Deviation	Initiating event on the model	First alarms (mm:ss)	SIS trip (first-out)	Range in the run
D1	N1 (N1.1)	Flow NO	Running sulfur pump P-101A trips and spare P-101B does not start	• FIC-103 LO at 00:01 • AIC-104 LO at 00:04 • TIC-201 LO at 00:06 • FI-100 LO at 00:14 • TIC-231 LO at 03:11 • +3 more	no trip in 30 min	• FI-103 0.0 to 40.9 t/h • TI-203 411 to 631 °C • TI-233 293 to 441 °C
D2	N2 (N2.1)	Flow NO	Main air blower K-101 trips	• FI-100 LO at 00:01 • AIT-306 HI at 00:01 • TIC-201 HI at 00:02 • AIC-104 HI at 00:03 • AI-105 LO at 00:03	FSLL-100 low total air at 00:06	• FI-100 0.0 to 238.5 kNm^3/h • FI-103 0.0 to 40.9 t/h • TI-203 631 to 631 °C
D3	N4 (N4.1)	Temperature MORE	WHB bypass TV-201 fails wide open	• TIC-201 HI at 00:03 • TI-203 HI at 02:05	TSHH-203 bed 1 exit high-high at 02:38	• TI-201 332 to 718 °C • TI-203 631 to 709 °C • AIT-306 5 to 292 ppm
D4	N2 (N2.4)	Flow LESS	FIC-101 put in MAN at 35 % (combustion air below the ratio)	• AIT-306 HI at 00:11 • AIC-104 HI at 00:14 • AI-105 LO at 00:18	no trip in 30 min	• FI-101 150.1 to 204.5 kNm^3/h • AI-105 7.57 to 9.00 % O_2 • AIT-306 292 to 1237 ppm
D5	N5 (N5.2)	Measurement OTHER THAN	TI-203 element failed low (open circuit) and TV-201 fails wide open	• TIC-201 HI at 00:03 • TIC-231 HI at 04:21 • TIC-211 HI at 06:09 • TIC-221 HI at 08:34 • TIC-241 HI at 12:26 • +2 more	ASHH-306 stack SO2 high-high at 26:24	• TI-202 628 to 792 °C • TI-203 25 to 631 °C • AIT-306 103 to 5000 ppm
D6	N2 (N2.5)	Flow LESS	Dilution air lost: FIC-102 in MAN at 0 % (AIC-104 in MAN)	• AIT-306 HI at 00:24 • AIC-104 HI at 00:26 • AI-105 LO at 00:33	ASHH-306 stack SO2 high-high at 07:00	• AI-104 0.00 to 14.00 % SO_2 • AI-105 7.00 to 21.00 % O_2 • AIT-306 72 to 4121 ppm • TI-203 627 to 631 °C
D7	N6 (N6.1)	Flow NO	IPA pump P-301A trips (HS-301B auto-start armed)	FI-301 LO at 00:02	no trip in 30 min	• FI-301 368 to 1600 m^3/h • AIT-306 292 to 486 ppm • AIT-307 10 to 26 mg/Nm^3
D8	N8 (N8.1)	Flow NO	FAT pump P-302A trips (P-302B manual spare)	• AIT-307 HI at 00:02 • FI-302 LO at 00:02	FSLL-302 FAT acid circulation low at 00:14	• FI-302 0 to 1000 m^3/h • AIT-307 10 to 3548 mg/Nm^3 • AIT-306 5 to 292 ppm
D9	N7 (N7.6)	Catalyst activity LESS	Bed-4 catalyst loses 90 % of its activity (10-min ramp)	AIT-306 HI at 03:51	no trip in 30 min	• TI-233 428 to 442 °C • XI-308 98.12 to 99.80 % • AIT-306 292 to 2737 ppm
D10	N9 (N9.1)	Utility NO	Instrument-air failure (every valve to its fail position)	• FIC-103 LO at 00:02 • TIC-201 LO at 00:03 • TIC-211 HI at 00:03 • TIC-251 HI at 00:04 • AIC-104 LO at 00:05 • +5 more	no trip in 30 min	• FI-103 0.0 to 40.9 t/h • FI-100 238.5 to 266.3 kNm^3/h • TI-203 221 to 631 °C

What the runs tell the HAZOP team

- **N2 blower trip (D2):** FSLL-100 trips the sulfur at 00:06 after K-101 stops. The trip works, but I-1 reads the same flow element as the FI-100 indication, so a meter fault defeats both (R08).
- **N1 loss of sulfur (D1):** no trip, because losing sulfur is the safe direction for the SIS. The air keeps sweeping the converter at the ratio floor of 100 kNm^3/h and in 30 min TI-203 falls to 411 °C and TI-233 to 293 °C, below the ≈355 °C strike of the Cs beds. The consequence is operational (a cold restart); a spare-pump auto-start (R01) removes most of it.
- **N4 TV-201 wide open (D3):** TI-203 alarms at 02:05 and TSHH-203 trips at 02:38 (33 s after the alarm). After the trip TI-203 keeps rising to 709 °C: the hot furnace refractory still heats the air and the open bypass sends it to bed 1. I-2 stops the sulfur but nothing closes TV-201 (R43).
- **N5 TI-203 failed low (D5):** the same TV-201 failure now gives no TI-203 alarm and no TSHH-203 trip, because the indication, the alarm and the trip share one element. Bed-1 middle TI-202 reaches 792 °C and the sulfur is only stopped at 26:24 by ASHH-306, after the stack SO_2 has been far above the permit (R20).
- **N2 dilution air lost (D6):** AI-104 rises to 14.0 % SO_2 and AIT-306 to 4121 ppm while TI-203 stays at or below 631 °C: the O_2 shortage limits conversion instead of overheating bed 1. ASHH-306 trips at 07:00, its 5-min delay included (R33).
- **N2 combustion air below the ratio (D4):** AI-105 falls to 7.6 % and AIT-306 peaks at 1237 ppm, ending at 388 ppm. No trip: only alarms protect (R04, R10).
- **N6/N8 acid pump trips (D7, D8):** with the HS-301B auto-start armed, FI-301 dips to 368 m^3/h and P-301B starts before FSLL-301 times out (no trip, AIT-307 peak 26 mg/Nm^3). The FAT spare has no auto-start: FSLL-302 trips at 00:14, but AIT-307 has already reached 3548 mg/Nm^3. The trip limits how long the mist lasts, not its peak (R24).
- **N7 bed-4 activity loss (D9):** AIT-306 alarms at 03:51 and climbs to 2737 ppm with no trip, because ASHH-306 acts only above 3000 ppm: a long permit exceedance is possible with every safeguard working (R33).
- **N9 instrument-air failure (D10):** every valve goes to its fail position, FV-103 stops the sulfur and the air rises to 266 kNm^3/h with no SIS trip; in 30 min TI-203 falls to 221 °C. Safe for people, costly in catalyst heat and restart time (R35).

5.5 Follow-up

- Each recommendation gets an owner and a due date; the owner answers *accepted and done*, *accepted with an alternative* (with justification) or *rejected* (with a justification agreed by the HAZOP leader and the plant manager), on the close-out sheet in Appendix 3.
- High-ranked scenarios protected by instrumented trips (bed-1 over-temperature with TSHH-203, loss of air with FSLL-100, loss of FAT acid with FSLL-302) go to LOPA to set the required SIL (IEC 61511).
- Changes made to close actions (new trip elements, auto-starts, cross-limiting, new alarms) go through management of change, which decides whether this HAZOP, the C&E matrix and the SRS must be revised.
- Revalidate the HAZOP periodically (typically every five years), after incidents and after a catalyst change or a capacity increase.
- Train operators on the simulator on the scenarios found here and verify that procedures and alarm responses work.

Appendix 1 HAZOP worksheets

N1 · Molten sulfur supply: P-101A/B → filter → XV-103 → FV-103 → F-101 burner

Project	SO3 Converter Ver 1 · converter section	**Drawing**	SO3-V1-PID-001 rev A (screen P&ID); SO3-PID-001 / -002 rev A (EPC)
Node	N1 · Molten sulfur supply: P-101A/B → filter → XV-103 → FV-103 → F-101 burner	**Design conditions**	40.94 t/h; ≈140 °C; supply pressure at FV-103 ≈ 600 kPa (model K.Psup); FV-103 at 50 %
Design intent	Deliver 40.94 t/h of molten sulfur at about 140 °C from the sulfur pit (outside the battery limit) with P-101A (P-101B spare) through the steam-jacketed filter, the SIS shut-off XV-103 and the flow valve FV-103 (FIC-103, fail closed) to the burner of F-101.	**Equipment**	P-101A/B, sulfur filter, XV-103, FV-103, steam-jacketed line DN80-MS-105
		Instruments	FIC-103 (FI-103), PDG-103 (local), XV-103 (SIS final element)

ID	Parameter	Guide word	Deviation	Causes	Consequences (no credit for safeguards)	Existing safeguards	S	L	Risk	Rec.	Action party
N1.1	Flow	**NO**	No sulfur to the burner	• Running pump P-101A trips (motor, power) • spare P-101B fails to start • XV-103 closed by a spurious SIS trip • FV-103 closes on instrument-air loss (fail closed) • sulfur frozen in the line	Furnace and converter cool; bed 1 falls toward its strike temperature; if not restored, a preheat with H-101 is needed; production loss	• FIC-103 alarm LO 35 t/h • AIC-104 alarm LO 11 % SO_2 • spare P-101B (manual start) • hot standby procedure	2	3	**Medium 6**	R01	Instrument / Operations
N1.2	Flow	**LESS**	Less sulfur than set	• Sulfur filter blinding • pump wear • FV-103 sticking • sulfur too viscous (cold or overheated) • low pit level	Lower SO_2 strength and production; the beds cool slowly because FFC-101 follows the measured sulfur	• FIC-103 alarm LO 35 t/h • FIC-103 opens FV-103 • PDG-103 local ΔP gauge on the filter • FFC-101 ratio 4.994 kNm^3/t sets the FIC-101 set point (floor 100 kNm^3/h)	2	3	**Medium 6**	R03, R02	Instrument / Mechanical / Process / Operations
N1.3	Flow	**MORE**	More sulfur than intended	• FIC-103 set point raised beyond design (production push) • FV-103 sticks open • FI-103 reads low	More SO_2 and heat; if the air does not follow (ratio station in MAN, FIC-101 at its limit) sulfur vapour leaves the furnace unburned and burns on bed 1: bed-1 overheating, catalyst sintering, stack SO_2 up	• FFC-101 ratio 4.994 kNm^3/t sets the FIC-101 set point (floor 100 kNm^3/h) • FIC-103 alarm HI 44 t/h • TI-203 alarm HI 640 °C • I-2 TSHH-203 trip at bed-1 exit > 660 °C, 5 s (same element as TI-203) • AIT-306 alarm HI 500 ppm	3	3	**Medium 9**	R04, R05	Instrument
N1.4	Temperature	**LESS**	Sulfur too cold	• Steam jacket or tracing failure • low jacket steam pressure • long stop with the line full	Sulfur freezes in the line, filter and burner gun: flow loss and plugging; a line can split when it is re-melted (expansion), releasing molten sulfur: burns, fire, SO_2	• Steam-jacketed line DN80-MS-105 • operator rounds (no temperature indication in the DCS)	3	3	**Medium 9**	R02	Process / Operations
N1.5	Temperature	**MORE**	Sulfur overheated (above about 160 °C)	Jacket steam at too high a pressure	Viscosity rises steeply: flow falls, FV-103 opens wide; more H_2S released from the sulfur (pit area, outside the battery limit)	• Jacket steam pressure set at the source (outside the battery limit, assumed) • FIC-103 alarm LO 35 t/h	2	2	**Low 4**	R02	Process / Operations
N1.6	Pressure	**MORE**	Pump discharge pressure high (dead-head)	• XV-103 or FV-103 closed with the pump running • filter blocked • burner gun plugged	Pump overheats at no flow; seal failure; molten sulfur leak at the pump: burns, sulfur fire, SO_2	• Pump shut-off head below the line design pressure (assumed) • operator • P-101A/B stop from the DCS	3	3	**Medium 9**	R06	Process / Instrument
N1.7	Pressure	**LESS**	Pump suction pressure low	• Low sulfur pit level • suction strainer plugged • suction line partly frozen	Cavitation, pump damage, loss of flow (see N1.1)	• FIC-103 alarm LO 35 t/h • pit level (outside the battery limit)	2	3	**Medium 6**	R01, R06	Instrument / Operations / Process

ID	Parameter	Guide word	Deviation	Causes	Consequences (no credit for safeguards)	Existing safeguards	S	L	Risk	Rec.	Action party
N1.8	Flow	**REVERSE**	Hot furnace gas back into the sulfur line	• Both pumps stopped with XV-103 open • discharge check valve passing	Hot SO_2 gas flows back toward the pumps and the pit: toxic release in the sulfur area, sulfur fire in the line	• XV-103 closes on every SIS trip • FV-103 fails closed • check valves on the pump discharge (assumed)	4	2	**Medium 8**	R06	Process / Instrument
N1.9	Integrity	**AS WELL AS**	Molten sulfur leak	• Pump seal failure • steam-jacket leak • filter opened while hot or under pressure during a change-over	Molten sulfur at 140 °C: burns; sulfur fire producing SO_2; possible H_2S	• XV-103 remote isolation (SIS) • pump stop from the DCS • I-6 HS ESD push-button (XV-103 closed, K-101 and H-101 stopped) • PPE	3	3	**Medium 9**	R07, R16	Operations / Safety
N1.10	Maintenance	**OTHER THAN**	Filter or pump change-over done incorrectly	• Spare filter not preheated (sulfur freezes) • valve line-up wrong • running pump stopped before the spare runs	Flow interruption (see N1.1); sulfur spill at the filter	• Procedure • FIC-103 alarm LO 35 t/h	2	3	**Medium 6**	R07	Operations
N1.11	Composition	**AS WELL AS**	Ash, carbon or water in the sulfur	• Poor sulfur quality • filter by-passed or element torn	Ash and carbon carried to the furnace and the first catalyst bed: burner and converter fouling, higher pressure drop over time; carbon burns and uses O_2; water forms acid on cold surfaces	• Sulfur filter • sulfur purchase specification (assumed) • PDI-204 alarm HI 8 kPa	2	3	**Medium 6**	R03, R21	Instrument / Mechanical / Process

N2 · Air supply K-101 and air split FV-101 (combustion) / FV-102 (dilution)

Project	SO3 Converter Ver 1 · converter section	**Drawing**	SO3-V1-PID-001 rev A (screen P&ID); SO3-PID-001 / -002 rev A (EPC)
Node	N2 · Air supply K-101 and air split FV-101 (combustion) / FV-102 (dilution)	**Design conditions**	238.5 kNm^3/h total; 90 °C; PI-101 60 kPa g; FV-101 55 %, FV-102 45 %; blower ≈4.2 MW
Design intent	K-101 delivers 238.5 kNm^3/h of dried air at 90 °C (PI-101 60 kPa g, ≈4.2 MW). FV-101 (fail open, FIC-101 in CAS from the FFC-101 ratio 4.994 kNm^3/t) sends 204.4 kNm^3/h to the furnace; FV-102 (fail open, FIC-102 in CAS from AIC-104) sends 34.1 kNm^3/h of dilution air to the bed-1 inlet.	**Equipment**	K-101 main air blower, FV-101, FV-102, air ducts from the drying tower (outside the battery limit)
		Instruments	FI-100 (FT-100) / FSLL-100, PI-101, FFC-101, FIC-101 (FI-101), FIC-102 (FI-102)

ID	Parameter	Guide word	Deviation	Causes	Consequences (no credit for safeguards)	Existing safeguards	S	L	Risk	Rec.	Action party
N2.1	Flow	**NO**	No air (blower stops)	• K-101 motor trip (power dip, motor protection) • blower suction blocked (drying tower, outside the battery limit)	Sulfur keeps flowing into a hot furnace without air: sulfur vapour accumulates in F-101, B-101 and the duct and can ignite violently when air returns; no gas through the beds	• I-1 FSLL-100 trip (XV-103) at total air < 90 kNm^3/h, 3 s • XV-103 closes • FI-100 alarm LO 200 kNm^3/h • I-6 HS ESD push-button (XV-103 closed, K-101 and H-101 stopped)	4	3	**High 12**	R08	Instrument
N2.2	Flow	**LESS**	Total air low	• Blower degradation • hot afternoon (less mass at the same valve openings) • higher system resistance (bed pressure drop, drying tower)	O_2 short in the furnace and the converter: conversion falls, stack SO_2 rises; at the limit FSLL-100 trips	• FI-100 alarm LO 200 kNm^3/h • AI-105 alarm LO 8 % O_2 • AIT-306 alarm HI 500 ppm • I-1 FSLL-100 trip (XV-103) at total air < 90 kNm^3/h, 3 s	2	4	**Medium 8**	R09	Process
N2.3	Flow	**MORE**	Total air high	• FV-101 and FV-102 fail open (instrument-air loss) • wrong FIC-101 set point or ratio entered	Furnace cooler, SO_2 strength lower, more cold air into bed 1: the beds can fall toward their strike temperature; blower motor overload	• FI-100 alarm HI 265 kNm^3/h • AIC-104 alarm LO 11 % SO_2 • TIC-201 alarm LO 405 / HI 435 °C	2	3	**Medium 6**	R12	Mechanical / Electrical
N2.4	Flow	**LESS**	Combustion air low against the sulfur	• FIC-101 in MAN at a low output • FI-101 reading high • FFC-101 ratio set low	O_2 in the furnace falls; unburned sulfur vapour carried forward burns on bed 1 or deposits in B-101; lower conversion	• FFC-101 ratio 4.994 kNm^3/t sets the FIC-101 set point (floor 100 kNm^3/h) • AI-105 alarm LO 8 % O_2 • AIC-104 alarm HI 12.8 % SO_2 • AIT-306 alarm HI 500 ppm	3	3	**Medium 9**	R04, R10	Instrument / Process
N2.5	Flow	**LESS**	Dilution air low or lost	• FIC-102 in MAN at a low output • AIC-104 driving FV-102 closed on a wrong analyser value • FV-102 actuator fault	SO_2 into bed 1 above 12 % with O_2 below 9 %: the O_2/SO_2 ratio limits conversion, stack SO_2 rises several times above the permit (see D6)	• AIC-104 alarm HI 12.8 % SO_2 • AI-105 alarm LO 8 % O_2 • AIT-306 alarm HI 500 ppm • I-5 ASHH-306 trip at stack SO_2 > 3000 ppm for 5 min (inhibited 60 min after admission)	3	3	**Medium 9**	R10, R33	Process / Instrument / Environment
N2.6	Flow	**MORE**	Dilution air high	• FV-102 fails open (instrument-air loss) • AIC-104 set point lowered • analyser reading high	Weaker gas and a colder bed-1 inlet: less heat for the beds, the second pass can fall toward its strike temperature; production loss	• AIC-104 alarm LO 11 % SO_2 • TIC-201 alarm LO 405 / HI 435 °C • TIC-231 alarm LO 405 / HI 445 °C	2	3	**Medium 6**	R11	Instrument / Operations
N2.7	Pressure	**MORE**	Blower discharge pressure high (surge)	• FV-101 and FV-102 closing together • high bed pressure drop • blocked duct	Blower surge, vibration and damage; duct overpressure, SO_2 leaks at flanges downstream	• FV-101/FV-102 fail open • PI-200 alarm HI 42 kPa g • PI-101 indication	3	2	**Medium 6**	R12	Mechanical / Electrical
N2.8	Temperature	**MORE**	Air at the blower hotter than design	• Hot afternoon • drying-tower acid hot (outside the battery limit)	Less mass at the same volume: lower O_2 and a slightly hotter bed-1 inlet	• FIC-101 and FIC-102 flow control • AI-105 alarm LO 8 % O_2	1	4	**Low 4**	R09	Process
N2.9	Composition	**AS WELL AS**	Moisture in the air	• Drying-tower upset (weak drying acid, low circulation • outside the battery limit)	Water forms sulfuric-acid mist with SO_3: corrosion of ducts, exchangers and the stack, acid-mist emission	• AIT-307 alarm HI 40 mg/Nm^3 • drying-tower instruments (outside the battery limit)	3	3	**Medium 9**	R13	Process

ID	Parameter	Guide word	Deviation	Causes	Consequences (no credit for safeguards)	Existing safeguards	S	L	Risk	Rec.	Action party
N2.10	Flow	**REVERSE**	Hot SO_2 gas back through the blower	• K-101 stops while the converter system is under pressure • no non-return damper	Hot SO_2-laden gas flows back to the blower and the drying tower: toxic exposure in the blower house	I-1 FSLL-100 trip (XV-103) at total air < 90 kNm^3/h, 3 s stops the sulfur, so no new SO_2 is made	4	2	**Medium 8**	R12	Mechanical / Electrical

N3 · Sulfur furnace F-101 with preheat burner H-101

Project	SO3 Converter Ver 1 · converter section	**Drawing**	SO3-V1-PID-001 rev A (screen P&ID); SO3-PID-001 / -002 rev A (EPC)
Node	N3 · Sulfur furnace F-101 with preheat burner H-101	**Design conditions**	1121 °C; 14 % SO_2, 7 % O_2; 105.3 MW; H-101 off in normal operation
Design intent	Burn 40.94 t/h of sulfur in 204.4 kNm^3/h of air to SO_2 gas at 1121 °C, 14 % SO_2, 7 % O_2 (105.3 MW). H-101 (0–30 MW, HIC-109) heats the furnace and the converter during start-up only.	**Equipment**	F-101 sulfur furnace (refractory lined), sulfur burner gun, H-101 preheat burner
		Instruments	TI-106, HIC-109, FI-109, H-101 flame detection (burner management)

ID	Parameter	Guide word	Deviation	Causes	Consequences (no credit for safeguards)	Existing safeguards	S	L	Risk	Rec.	Action party
N3.1	Temperature	**MORE**	Furnace too hot	• Air/sulfur ratio low (less excess air) • production push • H-101 left firing with sulfur in	Refractory damage, shell hot spots; hotter gas to B-101: tube-sheet overheating	• TI-106 indication (no alarm in the alarm list) • FFC-101 ratio 4.994 kNm^3/t sets the FIC-101 set point (floor 100 kNm^3/h) • TI-107 alarm HI 340 °C	3	2	**Medium 6**	R14	Process / Instrument
N3.2	Temperature	**LESS**	Furnace below sulfur ignition	• Sulfur admitted into a furnace that is not hot enough • a cold furnace after a long stop • large excess air	Sulfur does not ignite and pools in the furnace; the pool later ignites at once: temperature surge, sulfur vapour carry-over, SO_2 puff, overpressure	• start permissive TI-106 ≥ 450 °C • start permissive FIC-103 in MAN ≤ 25 % • nothing acts once sulfur is in	4	3	**High 12**	R14, R39	Process / Instrument / Operations
N3.3	Flame	**NO**	H-101 flame failure during preheat	• Fuel supply lost • flame-out at low air • burner fault	Unburned fuel enters the hot furnace; ignition of the accumulated fuel: explosion in F-101 or B-101	• H-101 own flame protection (fuel shut off) • H-101 trips itself when total air < 60 kNm^3/h • purge before relight	5	2	**High 10**	R15	Instrument
N3.4	Flow	**MORE**	H-101 firing too high	HIC-109 set high for a fast heat-up	Thermal shock to refractory and catalyst; bed 1 heated above 440 °C before sulfur	• start permissive TIC-201 and TI-202 ≥ 400 °C, inlet ≤ 440 °C • TIC-201 alarm HI 435 °C	2	3	**Medium 6**	R39	Operations
N3.5	Sequence	**AS WELL AS**	H-101 still firing after sulfur admission	Start-up step missed	Extra heat and combustion water (acid mist, corrosion); O_2 consumed; furnace hotter	• Procedure step • FI-109 indication	2	3	**Medium 6**	R39	Operations
N3.6	Composition	**PART OF**	Incomplete sulfur combustion	• O_2 short (see N2.4) • poor atomisation at the burner gun	Sulfur deposits in B-101 and on the bed-1 screen; burning on bed 1 gives a hot spot	• AI-105 alarm LO 8 % O_2 • TI-203 alarm HI 640 °C • I-2 TSHH-203 trip at bed-1 exit > 660 °C, 5 s (same element as TI-203)	3	2	**Medium 6**	R10	Process / Instrument
N3.7	Pressure	**MORE**	Furnace pressure surge	• Delayed ignition of a sulfur pool • blocked downstream	Refractory and duct damage; SO_2 leaks at flanges and sight ports	PI-200 alarm HI 42 kPa g	3	2	**Medium 6**	R14, R16	Process / Instrument / Safety
N3.8	Integrity	**AS WELL AS**	SO_2 leak from the furnace shell	• Refractory failure (hot spot) • corrosion where the shell is below the acid dew point	SO_2 release in the unit: toxic exposure of people nearby	• Shell temperature survey (assumed) • no fixed SO_2 detection on the P&ID	4	2	**Medium 8**	R16	Safety
N3.9	Measurement	**OTHER THAN**	TI-106 reads wrong	Thermocouple burned out or drifted (single element)	Start permissive based on a wrong value: sulfur admitted into a furnace that is too cold, or a hot furnace not seen	None (single element)	3	2	**Medium 6**	R14	Process / Instrument

N4 · Waste heat boiler B-101, hot-gas bypass TV-201 and dilution to bed 1

Project	SO3 Converter Ver 1 · converter section	**Drawing**	SO3-V1-PID-001 rev A (screen P&ID); SO3-PID-001 / -002 rev A (EPC)
Node	N4 · Waste heat boiler B-101, hot-gas bypass TV-201 and dilution to bed 1	**Design conditions**	Gas 1121 → 274 °C; steam 4.0 MPa g, 94 t/h; bed-1 inlet 420 °C, 12.0 % SO_2, 9.0 % O_2; PI-200 35 kPa g; TV-201 35 %
Design intent	Cool the furnace gas from 1121 to 274 °C in fire-tube boiler B-101 (4.0 MPa g steam, 94 t/h, ≈45 MW). TV-201 (fail closed, TIC-201) bypasses hot gas, and the dilution air sets the bed-1 inlet at 420 °C, 12.0 % SO_2 / 9.0 % O_2, 238.5 kNm^3/h, PI-200 35 kPa g.	**Equipment**	B-101 fire-tube boiler (steam drum outside the battery limit), TV-201 hot-gas bypass, dilution-air mixing duct
		Instruments	TI-107, FI-108, TIC-201 (TI-201), AIC-104 (AI-104), AI-105, PI-200

ID	Parameter	Guide word	Deviation	Causes	Consequences (no credit for safeguards)	Existing safeguards	S	L	Risk	Rec.	Action party
N4.1	Temperature	**MORE**	Bed-1 inlet too hot	• TV-201 fails open or its seat leaks • TIC-201 set point too high • B-101 fouled (gas leaves hotter) • TIC-201 measurement error	Bed-1 exit above the catalyst limit: sintering and permanent activity loss, lower bed-1 conversion, converter internals and shell overheated	• TIC-201 alarm LO 405 / HI 435 °C • TI-203 alarm HI 640 °C • I-2 TSHH-203 trip at bed-1 exit > 660 °C, 5 s (same element as TI-203) • TV-201 fails closed on air loss	3	3	**Medium 9**	R43, R22	Process / Instrument
N4.2	Temperature	**LESS**	Bed-1 inlet too cold	• TV-201 fails closed (instrument-air loss) • TIC-201 measurement error • dilution air high • furnace cold	Bed 1 approaches its strike temperature (≈375 °C): conversion collapses, unconverted SO_2 to the stack	• TIC-201 alarm LO 405 / HI 435 °C • AIT-306 alarm HI 500 ppm • I-5 ASHH-306 trip at stack SO_2 > 3000 ppm for 5 min (inhibited 60 min after admission)	3	3	**Medium 9**	R20, R22	Instrument / Safety
N4.3	Flow	**NO**	No water on the boiler side	BFW pump failure or drum level control failure (outside the battery limit)	Dry fire tubes with 1121 °C gas: tube and tube-sheet failure; steam explosion risk when water returns; water and steam into the gas: acid corrosion downstream	• Steam-drum level protection (outside the battery limit, assumed) • TI-107 alarm HI 340 °C	4	2	**Medium 8**	R17	Process / Instrument
N4.4	Integrity	**AS WELL AS**	B-101 tube leak (water into the gas)	• Tube corrosion • thermal fatigue at the tube sheet	Steam in the process gas forms acid with SO_3 downstream: corrosion of exchangers and towers, weak acid, catalyst wetting	• Steam/BFW balance (assumed) • AIT-307 alarm HI 40 mg/Nm^3	3	2	**Medium 6**	R18	Mechanical / Process
N4.5	Pressure	**MORE**	Converter inlet pressure high	• Rising bed pressure drop (dust, catalyst fines) • blockage downstream • mist-eliminator plugging	The blower moves back on its curve: less air; duct and converter overpressure, leaks	• PI-200 alarm HI 42 kPa g • PDI-204 alarm HI 8 kPa • PI-101 indication	2	3	**Medium 6**	R21	Process
N4.6	Composition	**MORE**	SO_2 strength into bed 1 high	• Dilution air low (see N2.5) • SO_2 analyser error • sulfur surge	Larger bed-1 temperature rise; O_2/SO_2 ratio low; conversion falls and the stack SO_2 rises	• AIC-104 alarm HI 12.8 % SO_2 • AI-105 alarm LO 8 % O_2 • TI-203 alarm HI 640 °C • I-2 TSHH-203 trip at bed-1 exit > 660 °C, 5 s (same element as TI-203) • I-5 ASHH-306 trip at stack SO_2 > 3000 ppm for 5 min (inhibited 60 min after admission)	3	3	**Medium 9**	R19, R10	Instrument / Process
N4.7	Composition	**LESS**	O_2 into bed 1 low	• Combustion air low • SO_2 strength high	Conversion limited by O_2; stack SO_2 above the permit	• AI-105 alarm LO 8 % O_2 • AIT-306 alarm HI 500 ppm	2	3	**Medium 6**	R10	Process / Instrument
N4.8	Measurement	**OTHER THAN**	SO_2 analyser AI-104 wrong	• Calibration drift • sample line blocked with sulfur or acid • analyser power loss	AIC-104 moves FV-102 the wrong way: the real gas strength is away from 12 % (see N4.6 and N2.6)	• AIC-104 alarm LO 11 / HI 12.8 % SO_2 • AI-105 alarm LO 8 % O_2 • FIC-102 range 0–80 kNm^3/h	3	3	**Medium 9**	R19	Instrument
N4.9	Heat transfer	**LESS**	B-101 removes less heat	• Gas-side fouling • low drum level	TV-201 closes to compensate; once it is closed the bed-1 inlet rises with no control left	• TI-107 alarm HI 340 °C • TIC-201 alarm LO 405 / HI 435 °C	2	3	**Medium 6**	R18	Mechanical / Process

N5 · Converter first pass: beds 1–3 with SH-201, E-201 and E-202 hot sides

Project	SO3 Converter Ver 1 · converter section	**Drawing**	SO3-V1-PID-001 rev A (screen P&ID); SO3-PID-001 / -002 rev A (EPC)
Node	N5 · Converter first pass: beds 1–3 with SH-201, E-201 and E-202 hot sides	**Design conditions**	Bed exits 631 / 528 / 465 °C; cumulative conversion 63.1 / 88.1 / 95.83 %; bed ΔP 4.5 / 2.9 / 3.2 kPa
Design intent	Convert SO_2 to 95.83 % over beds 1–3 (V_2O_5–K_2SO_4, 33.6 / 37.4 / 45.1 t) with interpass cooling: bed 1 420 → 631 °C; SH-201 (gas 631 → 598 °C, steam to 292 °C, bypass TV-231 / TIC-231); E-201 hot side (bypass TV-211, TIC-211 bed-2 inlet 445 °C); bed 2 445 → 528 °C; E-202 hot side (bypass TV-221, TIC-221 bed-3 inlet 440 °C); bed 3 440 → 465 °C.	**Equipment**	R-201 beds 1–3, SH-201, E-201 (hot side), E-202 (hot side), TV-211, TV-221
		Instruments	TI-202, TI-203 / TSHH-203, PDI-204, TIC-211, TI-212, TI-213, PDI-214, TIC-221, TI-222, TI-223, PDI-224, TI-207, TI-253

ID	Parameter	Guide word	Deviation	Causes	Consequences (no credit for safeguards)	Existing safeguards	S	L	Risk	Rec.	Action party
N5.1	Temperature	**MORE**	Bed-1 exit above the catalyst limit	• Bed-1 inlet too hot (see N4.1) • SO_3 strength high • unburned sulfur burning on bed 1	Catalyst sintering and permanent activity loss; active-phase migration; grids and division plates distort; shell overstress	• TI-203 alarm HI 640 °C • I-2 TSHH-203 trip at bed-1 exit > 660 °C, 5 s (same element as TI-203) • TIC-201 alarm LO 405 / HI 435 °C	3	3	**Medium 9**	R20, R43	Instrument / Safety / Process
N5.2	Measurement	**OTHER THAN**	TI-203 element fails low	• Open-circuit thermocouple • wiring fault	The TI-203 alarm and the TSHH-203 trip read the same element, so both are blind: a real bed-1 over-temperature goes undetected (see D5: bed-1 middle near 790 °C, no TSHH-203 trip)	• TI-202 bed-1 middle indication (no alarm) • AIT-306 alarm HI 500 ppm late	4	3	**High 12**	R20	Instrument / Safety
N5.3	Temperature	**LESS**	Bed-2 or bed-3 inlet too cold	• TV-211 or TV-221 stuck or closed • TIC-211/TIC-221 set point low • bypass valve sticking	Bed approaches its strike temperature; conversion falls, stack SO_2 rises	• TIC-211 alarm LO 430 / HI 460 °C • TIC-221 alarm LO 425 / HI 455 °C • AIT-306 alarm HI 500 ppm	2	3	**Medium 6**	R22	Instrument
N5.4	Temperature	**MORE**	Bed-2 or bed-3 inlet too hot	• TV-211 / TV-221 fail open (instrument-air loss) • exchanger hot side fouled	Lower equilibrium conversion in beds 2–3; more SO_2 to the IPA and the second pass; stack SO_2 up	• TIC-211 alarm HI 460 °C • TIC-221 alarm HI 455 °C • AIT-306 alarm HI 500 ppm	2	3	**Medium 6**	R22	Instrument
N5.5	Control	**OTHER THAN**	A bed-inlet loop oscillates	• Bypass valve stick-slip • poor tuning • interaction between the bed-inlet loops through the return gas	Bed temperatures and conversion swing; thermal fatigue of exchangers	• TIC-211 alarm LO 430 / HI 460 °C • TIC-221 alarm LO 425 / HI 455 °C • tuning records	2	3	**Medium 6**	R22	Instrument
N5.6	Pressure	**MORE**	Bed pressure drop high	• Dust and ash on the bed screens • catalyst breakage from thermal cycling	Higher PI-200, less air from K-101; maldistribution and channelling give hot spots	• PDI-204 alarm HI 8 kPa • PDI-214, PDI-224 indications • PI-200 alarm HI 42 kPa g	2	3	**Medium 6**	R21	Process
N5.7	Flow	**PART OF**	Gas bypasses catalyst inside the converter	• Division-plate or bed-seal failure • settled catalyst leaving a gap at the wall	Part of the gas is not converted; overall conversion falls; stack SO_2 above the permit	• AIT-306 alarm HI 500 ppm • XI-308 calculated conversion • I-5 ASHH-306 trip at stack SO_2 > 3000 ppm for 5 min (inhibited 60 min after admission)	2	3	**Medium 6**	R18	Mechanical / Process
N5.8	Catalyst activity	**LESS**	Catalyst activity low in beds 1–3	• Ageing • dust • poisoning (arsenic, chlorides, water) • past overheating	Less conversion per bed, more work left to the later beds; stack SO_2 up; early catalyst replacement	• Bed temperature indications • AIT-306 alarm HI 500 ppm	2	3	**Medium 6**	R21	Process
N5.9	Integrity	**AS WELL AS**	E-201 or E-202 tube leak	• Acid corrosion at the cold end • thermal fatigue • tube-to-tubesheet weld failure	First-pass gas leaks into the return gas and skips the IPA: overall conversion falls, stack SO_2 and acid mist rise; corrosion	• AIT-306 alarm HI 500 ppm • I-5 ASHH-306 trip at stack SO_2 > 3000 ppm for 5 min (inhibited 60 min after admission)	3	3	**Medium 9**	R18	Mechanical / Process

ID	Parameter	Guide word	Deviation	Causes	Consequences (no credit for safeguards)	Existing safeguards	S	L	Risk	Rec.	Action party
N5.10	Temperature	**MORE**	Superheated steam too hot	• Low steam flow through SH-201 (less steam from B-101) • TV-231 closed	Superheater tube overheating; damage to the steam user (outside the battery limit)	• TI-207 indication (no alarm) • TIC-231 bypass	3	2	**Medium 6**	R23	Instrument

N6 · Economizer E-203, intermediate absorber T-301 and IPA acid circulation

Project	SO3 Converter Ver 1 · converter section	**Drawing**	SO3-V1-PID-001 rev A (screen P&ID); SO3-PID-001 / -002 rev A (EPC)
Node	N6 · Economizer E-203, intermediate absorber T-301 and IPA acid circulation	**Design conditions**	Gas to T-301 190 °C, out 83 °C; acid 1600 m^3/h, 98.5 %, 80 °C; BFW 150 t/h
Design intent	Cool the bed-3 gas to 190 °C in economizer E-203 (bypass TV-251, TIC-251) and absorb the SO_3 in T-301 with 1600 m^3/h of 98.5 % acid at 80 °C (P-301A, P-301B spare with the HS-301B auto-start on FSLL-301); the gas leaves at 83 °C to the E-202 cold side.	**Equipment**	E-203, TV-251, T-301 IPA, P-301A/B, acid pump tank TK-301, acid cooler (outside the model)
		Instruments	TIC-251 (TI-251), FI-253, TI-255, FI-301 / FSLL-301, HS-301B, AI-303, TI-304, TI-305

ID	Parameter	Guide word	Deviation	Causes	Consequences (no credit for safeguards)	Existing safeguards	S	L	Risk	Rec.	Action party
N6.1	Flow	**NO**	No IPA acid circulation	• P-301A trips and the P-301B auto-start fails • suction valve closed • pump tank empty	SO_3 passes the tower unabsorbed: dense acid mist from the stack and SO_3 into the second pass; hot dry gas damages the tower lining; stack SO_2 up (see D7, D8)	• HS-301B auto-start of P-301B • FI-301 alarm LO 1200 m^3/h • I-3 FSLL-301 trip at IPA acid < 800 m^3/h, 10 s	4	2	**Medium 8**	R24, R25	Instrument / Operations
N6.2	Flow	**LESS**	IPA acid flow low	• Pump wear • strainer or distributor plugging	Poorer absorption, acid mist; dry spots in the packing	• FI-301 alarm LO 1200 m^3/h • I-3 FSLL-301 trip at IPA acid < 800 m^3/h, 10 s	2	3	**Medium 6**	R25	Instrument
N6.3	Composition	**LESS**	IPA acid strength low	• Dilution water valve passing • cross-over from another acid system • wrong sample	SO_3 absorption falls: acid mist carried forward and emitted; faster corrosion of acid piping (weaker acid)	• AI-303 alarm LO 98 % • AIT-307 alarm HI 40 mg/Nm^3	3	3	**Medium 9**	R26	Process
N6.4	Composition	**MORE**	IPA acid strength high	Dilution water lost	SO_3 slip over strong acid; acid mist	AI-303 indication (no high alarm)	2	2	**Low 4**	R23, R26	Instrument / Process
N6.5	Temperature	**MORE**	IPA acid too hot	• Acid cooler fouled • cooling water lost (outside the battery limit)	Absorption falls (mist); corrosion of acid piping and coolers rises sharply	TI-304 alarm HI 95 °C	2	3	**Medium 6**	R27	Process
N6.6	Temperature	**MORE**	Gas to T-301 too hot	• TV-251 fails open (instrument-air loss) • economizer BFW lost • E-203 fouled	Hot gas into the tower: lining and packing damage, acid evaporation and mist	• TIC-251 alarm LO 165 / HI 230 °C • FI-253 alarm LO 110 t/h	2	3	**Medium 6**	R23, R17	Instrument / Process
N6.7	Temperature	**LESS**	Gas to T-301 too cold	• TV-251 closed • low gas flow • TIC-251 set point low	Metal below the acid dew point in E-203: condensation and corrosion; tube leak lets BFW into the gas	TIC-251 alarm LO 165 °C	3	2	**Medium 6**	R28	Process
N6.8	Integrity	**AS WELL AS**	Acid leak at the IPA pumps or piping	• Seal failure • corrosion • flange leak	98.5 % acid at 80 °C: severe chemical burns; spill	• I-6 HS ESD push-button (XV-103 closed, K-101 and H-101 stopped) • acid pumps stop from the DCS • PPE, safety showers (assumed)	4	3	**High 12**	R29, R40	Safety / Operations
N6.9	Integrity	**AS WELL AS**	E-203 tube leak (BFW into the gas)	• Cold-end corrosion • thermal fatigue	Water into SO_3 gas: acid in the duct and the IPA; corrosion; weak acid	• AI-303 alarm LO 98 % (late) • FI-253 against the steam balance	3	2	**Medium 6**	R18	Mechanical / Process
N6.10	Flow	**MORE**	IPA acid flow high	Both pumps running	Tower flooding; acid carry-over to the E-202 cold side: corrosion	FI-301 indication (no high alarm)	2	2	**Low 4**	R23	Instrument

N7 · Return gas (E-202 / E-201 cold sides), beds 4–5 and E-204

Project	SO3 Converter Ver 1 · converter section	**Drawing**	SO3-V1-PID-001 rev A (screen P&ID); SO3-PID-001 / -002 rev A (EPC)
Node	N7 · Return gas (E-202 / E-201 cold sides), beds 4–5 and E-204	**Design conditions**	Return gas 83 → 204 → 425 °C; bed 4 425 → 441 °C (99.70 %); bed 5 428 → 427 °C (99.80 %); strike ≈355 °C
Design intent	Reheat the absorbed gas from 83 to 204 °C in E-202 and to 425 °C in E-201 (the bed-4 inlet depends on the heat beds 1–2 give up, trimmed by TV-231 at SH-201); convert to 99.70 % in Cs-promoted bed 4 (425 → 441 °C) and 99.80 % in bed 5 (428 → 427 °C) after E-204 (bypass TV-241, TIC-241).	**Equipment**	E-202 and E-201 cold sides, R-201 beds 4–5 (Cs-promoted, 37.4 / 33.6 t), E-204, TV-241, HV-112 start-up line (joins at the bed-4 inlet)
		Instruments	TI-305, TI-254, TIC-231 (TI-231), TI-232, TI-233, PDI-234, TIC-241 (TI-241), TI-242, TI-243, PDI-244

ID	Parameter	Guide word	Deviation	Causes	Consequences (no credit for safeguards)	Existing safeguards	S	L	Risk	Rec.	Action party
N7.1	Temperature	**LESS**	Bed-4 inlet too cold	• Less heat available from beds 1–2 (low rate, weak gas) • TV-231 at its limit • interpass exchangers fouled or bypassing • IPA gas cold	Bed 4 approaches its strike temperature; final conversion falls; stack SO_2 above the permit	• TIC-231 alarm LO 405 / HI 445 °C • AIT-306 alarm HI 500 ppm • I-5 ASHH-306 trip at stack SO_2 > 3000 ppm for 5 min (inhibited 60 min after admission)	2	4	**Medium 8**	R30	Process
N7.2	Temperature	**MORE**	Bed-4 inlet too hot	• TV-231 fails open (instrument-air loss), so more heat reaches E-201 • high production	Lower equilibrium conversion in bed 4; stack SO_2 up	• TIC-231 alarm HI 445 °C • AIT-306 alarm HI 500 ppm	2	3	**Medium 6**	R22	Instrument
N7.3	Composition	**AS WELL AS**	Unconverted gas reaches the second pass	• Start-up line HV-112 left open or passing • interpass exchanger tube leak (see N5.9) • internal bypass in the first pass (see N5.7)	Final conversion falls; stack SO_2 rises above the 500 ppm permit for as long as it lasts	• HV-112 locked closed in operation • start permissive HV-112 closed • AIT-306 alarm HI 500 ppm • I-5 ASHH-306 trip at stack SO_2 > 3000 ppm for 5 min (inhibited 60 min after admission)	3	3	**Medium 9**	R31, R18	Operations / Instrument / Mechanical / Process
N7.4	Temperature	**LESS**	Bed-5 inlet too cold	• TV-241 closed or stuck • E-204 removing more heat than needed	Bed 5 toward its strike temperature; final conversion falls	• TIC-241 alarm LO 410 / HI 445 °C • AIT-306 alarm HI 500 ppm	2	3	**Medium 6**	R22	Instrument
N7.5	Temperature	**MORE**	Bed-5 inlet too hot	• TV-241 fails open • BFW to E-204 lost	Bed-5 equilibrium lower; stack SO_2 slightly up	• TIC-241 alarm HI 445 °C • FI-253 alarm LO 110 t/h	1	3	**Low 3**	—	—
N7.6	Catalyst activity	**LESS**	Bed-4 catalyst activity low	• Catalyst wetted by acid during a shutdown • ageing • dust	Stack SO_2 rises toward and above the permit but stays below the ASHH-306 trip (see D9): a long exceedance with no trip	• AIT-306 alarm HI 500 ppm • I-5 ASHH-306 trip at stack SO_2 > 3000 ppm for 5 min (inhibited 60 min after admission) only far above the permit	3	3	**Medium 9**	R21, R33	Process / Environment
N7.7	Pressure	**MORE**	Bed 4 / bed 5 pressure drop high	• Dust • sulfate crusts after acid wetting	Higher system pressure and less air	• PDI-234, PDI-244 indications • PI-200 alarm HI 42 kPa g	1	3	**Low 3**	R21	Process
N7.8	Integrity	**AS WELL AS**	Corrosion of the cold return-gas side	• Acid mist carried over from T-301 (mist eliminator damaged) • metal below the dew point at the E-202 cold end	Duct and E-202 tube corrosion; leaks	• T-301 mist eliminators (assumed) • TI-254 indication	2	3	**Medium 6**	R32	Mechanical

N8 · Economizer E-205, final absorber T-302, stack ST-301 and stack analysers

Project	SO3 Converter Ver 1 · converter section	**Drawing**	SO3-V1-PID-001 rev A (screen P&ID); SO3-PID-001 / -002 rev A (EPC)
Node	N8 · Economizer E-205, final absorber T-302, stack ST-301 and stack analysers	**Design conditions**	Gas to T-302 160 °C; acid 1000 m^3/h; stack 292 ppm SO_2, 10.5 mg/Nm^3 mist; 99.80 % conversion; 125 t/h H_2SO_4
Design intent	Cool the bed-5 gas to 160 °C in E-205 (no bypass) and absorb the remaining SO_3 in T-302 with 1000 m^3/h of acid (P-302A, P-302B manual spare); tail gas to the stack at 292 ppm SO_2 (permit 500 ppm, AIT-306) and 10.5 mg/Nm^3 acid mist (AIT-307); XI-308 99.80 %; FI-309 125 t/h acid.	**Equipment**	E-205, T-302 FAT, P-302A/B, acid pump tank TK-302, stack ST-301
		Instruments	TI-252, FI-302 / FSLL-302, AIT-306 / ASHH-306, AIT-307, XI-308, FI-309

ID	Parameter	Guide word	Deviation	Causes	Consequences (no credit for safeguards)	Existing safeguards	S	L	Risk	Rec.	Action party
N8.1	Flow	**NO**	No FAT acid circulation	• P-302A trips (P-302B is a manual spare) • suction blocked	SO_3 goes up the stack as a dense acid-mist plume that can reach people downwind (see D8: AIT-307 far above its alarm before the trip); tower lining damage	• FI-302 alarm LO 750 m^3/h • I-4 FSLL-302 trip at FAT acid < 500 m^3/h, 10 s • AIT-307 alarm HI 40 mg/Nm^3	4	3	**High 12**	R24	Instrument / Operations
N8.2	Flow	**LESS**	FAT acid flow low	• Pump wear • strainer or distributor plugging	Poorer final absorption; acid mist	• FI-302 alarm LO 750 m^3/h • I-4 FSLL-302 trip at FAT acid < 500 m^3/h, 10 s	2	3	**Medium 6**	R24	Instrument / Operations
N8.3	Composition	**MORE**	Stack SO_2 above the permit	• Any conversion loss upstream (bed temperatures, catalyst, bypassing, O_2 short) • start-up	Permit exceeded: fines, complaints; at high levels a health risk downwind	• AIT-306 alarm HI 500 ppm • I-5 ASHH-306 trip at stack SO_2 > 3000 ppm for 5 min (inhibited 60 min after admission)	3	4	**High 12**	R33, R34	Process / Environment / Instrument
N8.4	Composition	**MORE**	Stack acid mist high	• FAT acid strength or temperature off • candle filter failure • moisture in the air	Visible plume, local acid fallout, corrosion of the stack	AIT-307 alarm HI 40 mg/Nm^3	2	3	**Medium 6**	R32	Mechanical
N8.5	Temperature	**MORE**	Gas to T-302 too hot	• Economizer BFW lost • E-205 fouled	Poorer final absorption; tower lining damage	• TI-252 indication (no alarm) • FI-253 alarm LO 110 t/h	2	3	**Medium 6**	R23	Instrument
N8.6	Temperature	**LESS**	Gas to T-302 below the acid dew point	Low production rate with full BFW flow (E-205 has no bypass)	Acid condensation in E-205 and the duct: corrosion and tube leaks	None (E-205 has no bypass)	2	3	**Medium 6**	R28	Process
N8.7	Measurement	**OTHER THAN**	Stack SO_2 analyser AIT-306 wrong or frozen	• Sample line plugged • calibration drift • range limit (0–5000 ppm)	A permit exceedance is not seen; ASHH-306 reads the same analyser and does not trip	• XI-308 calculated conversion • analyser calibration (assumed)	3	3	**Medium 9**	R34	Instrument
N8.8	Measurement	**OTHER THAN**	Acid-mist analyser AIT-307 wrong	Sample line or optics fouled	A mist emission is not seen	Visual check of the plume	2	3	**Medium 6**	R34	Instrument
N8.9	Integrity	**AS WELL AS**	Acid leak at the FAT pumps or piping	• Seal failure • corrosion • flange leak	Concentrated acid: severe chemical burns; spill	• I-6 HS ESD push-button (XV-103 closed, K-101 and H-101 stopped) • pumps stop from the DCS • PPE, safety showers (assumed)	4	2	**Medium 8**	R29	Safety

N9 · Utilities: instrument air, power, BFW, steam and cooling water

Project	SO3 Converter Ver 1 · converter section	**Drawing**	SO3-V1-PID-001 rev A (screen P&ID); SO3-PID-001 / -002 rev A (EPC)
Node	N9 · Utilities: instrument air, power, BFW, steam and cooling water	**Design conditions**	BFW 150 t/h at 105 °C; steam 94 t/h; K-101 ≈4.2 MW
Design intent	Supply instrument air to every control valve (each goes to its fail position on loss), electric power to K-101 (≈4.2 MW) and the pumps, BFW 150 t/h to E-203 / E-204 / E-205 (FI-253), steam for the sulfur jackets and cooling water for the acid coolers; take 94 t/h of steam from B-101 through SH-201.	**Equipment**	Instrument-air header, electrical supply, BFW header, steam and cooling-water headers (all outside the battery limit)
		Instruments	FI-253, FI-108, TI-207; fail positions: FV-101, FV-102 open; FV-103 closed; TV-201 closed; TV-211 to TV-251 open; H-101 off

ID	Parameter	Guide word	Deviation	Causes	Consequences (no credit for safeguards)	Existing safeguards	S	L	Risk	Rec.	Action party
N9.1	Utility	**NO**	Instrument-air failure	• Compressor trip • dryer failure • header leak	Every valve to its fail position: FV-103 closes (sulfur lost), FV-101/FV-102 open (air to its maximum), TV-201 closes, TV-211 to TV-251 open, H-101 off; no SIS trip, the beds cool fast with full air (see D10)	• Fail-safe valve positions (sulfur off) • FIC-103 alarm LO 35 t/h • FI-100 alarm HI 265 kNm^3/h	2	3	**Medium 6**	R35	Instrument
N9.2	Utility	**NO**	Power failure	• Grid loss • switchboard fault	K-101 and all pumps stop; sulfur stops; acid circulation stops with hot gas in the towers; on restart, the blower before the acid pumps blows SO_3 to the stack	• I-1 FSLL-100 trip (XV-103) at total air < 90 kNm^3/h, 3 s • I-3 FSLL-301 trip at IPA acid < 800 m^3/h, 10 s • I-4 FSLL-302 trip at FAT acid < 500 m^3/h, 10 s • SIS de-energise to trip (assumed)	3	3	**Medium 9**	R36	Electrical / Operations
N9.3	Utility	**LESS**	Economizer BFW low	• BFW pump trip • control fault (outside the battery limit)	E-203 / E-204 / E-205 gas outlets hotter; steaming in the economizers (hammer, tube damage); hotter gas into the IPA and the FAT	• FI-253 alarm LO 110 t/h • TIC-251 alarm HI 230 °C	3	3	**Medium 9**	R17	Process / Instrument
N9.4	Utility	**NO**	Jacket steam lost	• Steam header fault • trap failures	Sulfur freezes in the jacketed line (see N1.4)	Operator rounds	2	3	**Medium 6**	R02	Process / Operations
N9.5	Utility	**MORE**	Steam header pressure high	Steam user trips (outside the battery limit)	Steam-drum safety valves lift; B-101 at higher pressure and temperature	Steam-drum safety valves (outside the battery limit, assumed)	2	2	**Low 4**	R17	Process / Instrument
N9.6	Utility	**NO**	Cooling water to the acid coolers lost	Cooling-water pump trip (outside the battery limit)	Acid temperature rises (see N6.5): absorption falls, corrosion rises	TI-304 alarm HI 95 °C	2	3	**Medium 6**	R27	Process
N9.7	Safety system	**NO**	XV-103 fails to close on demand	• Valve stuck (sulfur solidified on the seat) • solenoid or actuator fault	Every trip acts on this one valve: the hazard each trip protects against goes ahead	FV-103 closes as well, because the trip puts FIC-103 in MAN at 0 % (control system, not independent)	4	2	**Medium 8**	R37	Instrument / Safety

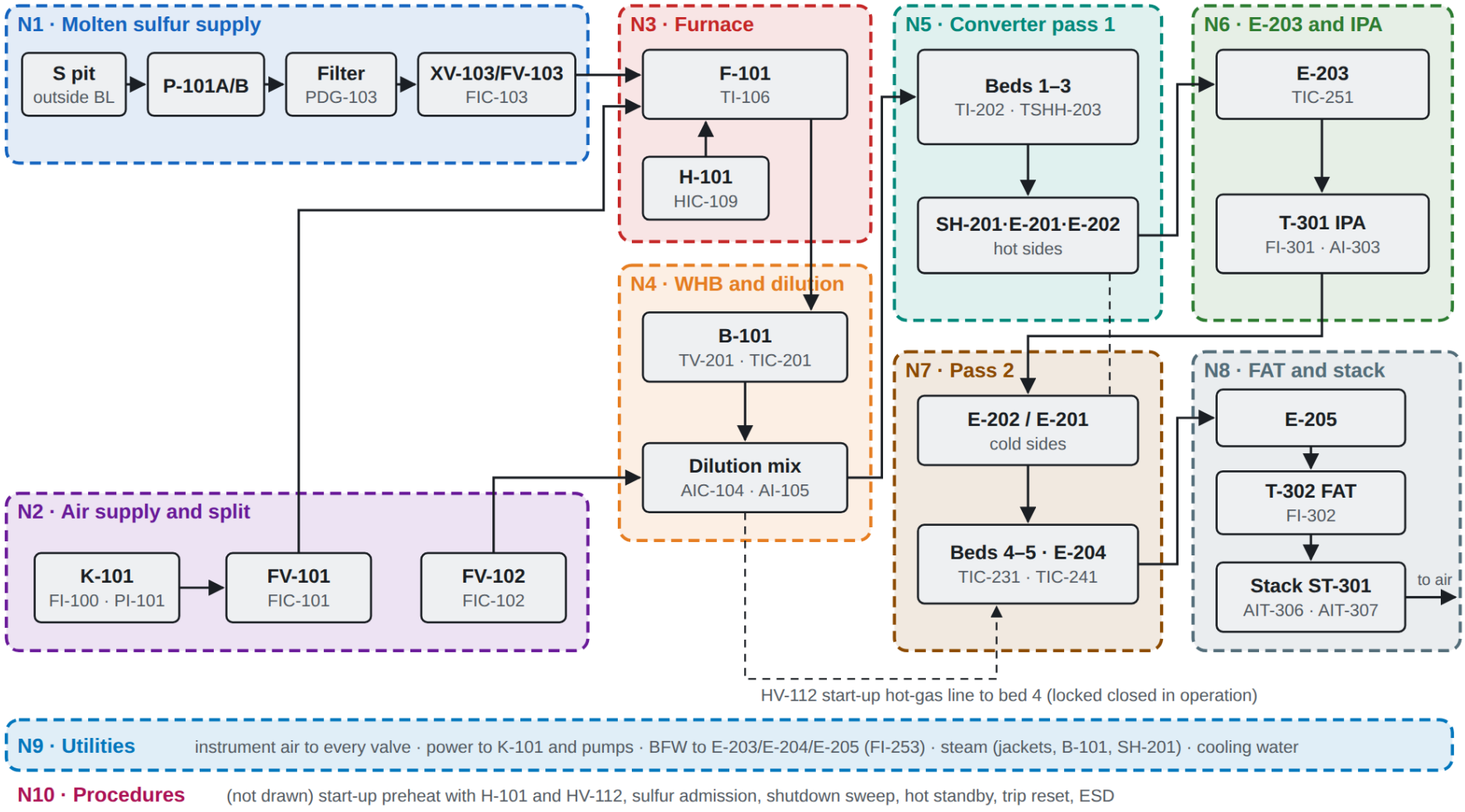
N10 · Procedures: start-up preheat, sulfur admission, shutdown sweep, hot standby, trip reset, ESD

Project	SO3 Converter Ver 1 · converter section	**Drawing**	SO3-V1-PID-001 rev A (screen P&ID); SO3-PID-001 / -002 rev A (EPC)
Node	N10 · Procedures: start-up preheat, sulfur admission, shutdown sweep, hot standby, trip reset, ESD	**Design conditions**	Cold start-up ≈4.5 h on the expert run; hot restart ≈1 h; shutdown ≈18 min; ESD 90 s (graded procedures)
Design intent	Preheat the furnace and the beds with H-101 (HV-112 open so hot gas also reaches beds 4–5), close HV-112, admit sulfur through the SIS when the start permissives hold, line out; stop sulfur, sweep 5 min at about 100 kNm^3/h and stop K-101 and the acid pumps to hold the beds hot (hot standby); emergency shutdown with the ESD push-button.	**Equipment**	H-101, HV-112, XV-103, K-101, P-301A/B, P-302A/B, SIS
		Instruments	Start permissives (SIS tab), first-out, HIC-109, TI-106, TI-202, TI-232, FI-301, FI-302, AIT-306

ID	Parameter	Guide word	Deviation	Causes	Consequences (no credit for safeguards)	Existing safeguards	S	L	Risk	Rec.	Action party
N10.1	Preheat	**LESS**	Beds not hot enough at sulfur admission	• Permissive bypassed • bed-4 thermocouple wrong • admission before the second pass is hot	SO_2 passes unconverted to the stack during the start-up (ASHH-306 is inhibited); permit exceedance	• start permissive TIC-201 and TI-202 ≥ 400 °C, inlet ≤ 440 °C • start permissive TI-232 ≥ 360 °C • AIT-306 alarm HI 500 ppm	2	3	**Medium 6**	R39, R42	Operations
N10.2	Sequence	**BEFORE**	Sulfur admitted before acid circulates	Steps taken out of order	SO_3 not absorbed: acid-mist plume	• start permissive FI-301 and FI-302 ≥ 80 % • I-3 FSLL-301 trip at IPA acid < 800 m^3/h, 10 s • I-4 FSLL-302 trip at FAT acid < 500 m^3/h, 10 s	3	2	**Medium 6**	R39	Operations
N10.3	Sequence	**AFTER**	HV-112 closed late or not at all	• Step missed • chain not fitted • valve not fully seated	Part of the converter feed skips beds 1–3 and the IPA: final conversion falls, stack SO_2 above the permit	• start permissive HV-112 closed • HV-112 locked closed	3	3	**Medium 9**	R31	Operations / Instrument
N10.4	Sulfur admission	**MORE**	Too much sulfur at admission	FIC-103 not in MAN or output above 25 %	Sulfur slug into a furnace just at ignition: pooling and puff; strong gas on beds that are not lined out	• start permissive FIC-103 in MAN ≤ 25 % • start permissive TI-106 ≥ 450 °C	3	2	**Medium 6**	R39	Operations
N10.5	Sweep	**LESS**	Shutdown sweep too short	Blower stopped straight after the sulfur stop	SO_2 and SO_3 left in the converter and ducts: released when equipment is opened; SO_3 condenses as acid when the converter cools (catalyst wetting)	Procedure: 5-min sweep at about 100 kNm^3/h with the stack SO_2 falling	3	3	**Medium 9**	R38	Operations
N10.6	Sweep	**MORE**	Air sweep too long	Blower left running after the sweep	Beds cool below their strike temperatures: a hot restart becomes a cold start-up	Procedure	1	4	**Low 4**	R38	Operations
N10.7	Hot standby	**LATE**	Restart after the beds have cooled	Repair takes longer than planned	Preheat with H-101 needed; risk of admission into beds below strike (see N10.1)	• start permissive TIC-201 and TI-202 ≥ 400 °C, inlet ≤ 440 °C • start permissive TI-232 ≥ 360 °C	1	3	**Low 3**	—	—
N10.8	Trip reset	**EARLY**	SIS reset and sulfur re-admitted before the cause is cleared	• Pressure to restart • first-out not reviewed	Repeat trip; sulfur into a furnace or beds that have cooled; over-temperature if the cause was a hot bed	• Reset blocked while TSHH-203 is active • start permissives at re-admission	3	2	**Medium 6**	R40	Operations
N10.9	ESD	**LATE**	Emergency shutdown too late in an acid or SO_2 leak	• Hesitation • no clear criteria	Leak continues: exposure of operators and responders	I-6 HS ESD push-button (XV-103 closed, K-101 and H-101 stopped)	4	2	**Medium 8**	R40, R29	Operations / Safety
N10.10	Maintenance	**OTHER THAN**	Equipment opened with SO_2/SO_3 or sulfur present	• Incomplete sweep • isolation incomplete (XV-103 passing, sulfur line not blinded)	Toxic exposure of the maintenance crew; molten sulfur release	Permit to work and gas test (assumed)	4	2	**Medium 8**	R41	Operations / Safety
N10.11	Safety system	**AS WELL AS**	A trip or permissive left bypassed	Bypass for maintenance or start-up not removed	The protection is missing when the demand comes (over-temperature, loss of air, acid loss)	Administrative control only	4	3	**High 12**	R42	Operations

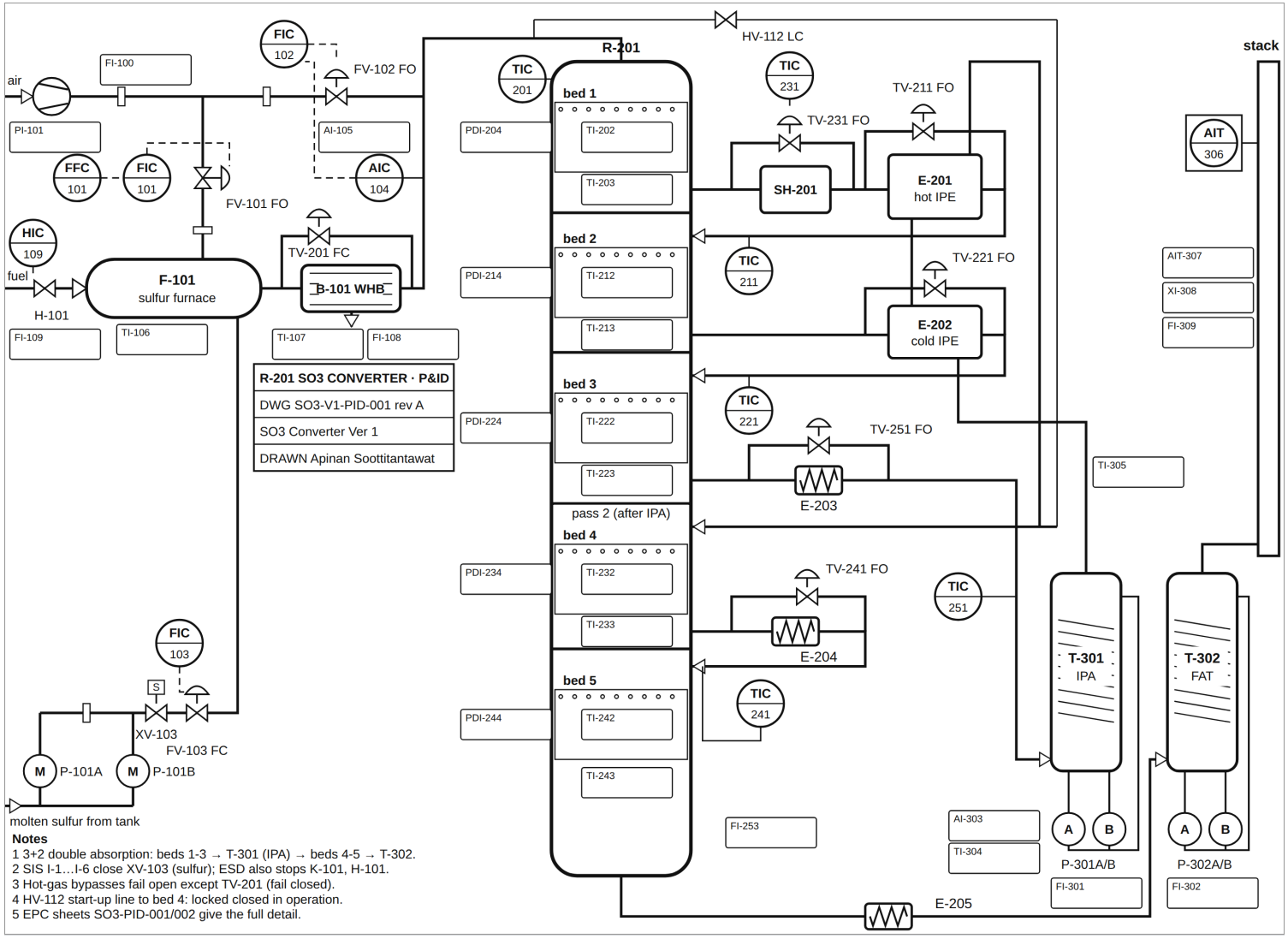
Appendix 2 Drawings

A2.1 HAZOP node map



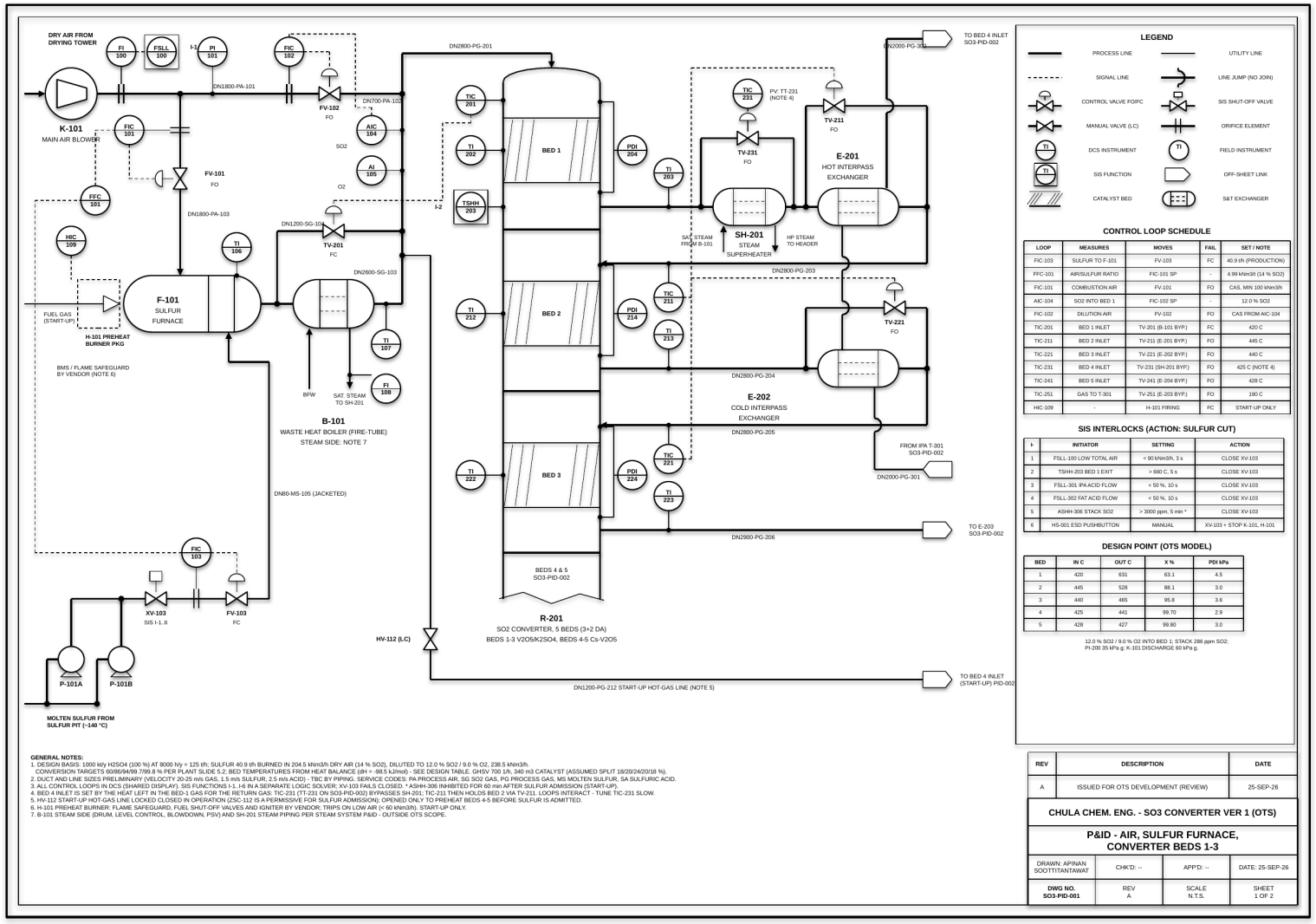
Drawing A2.1 — HAZOP node map N1 to N10 (from the HAZOP tab).

A2.2 Screen P&ID SO3-V1-PID-001 rev A



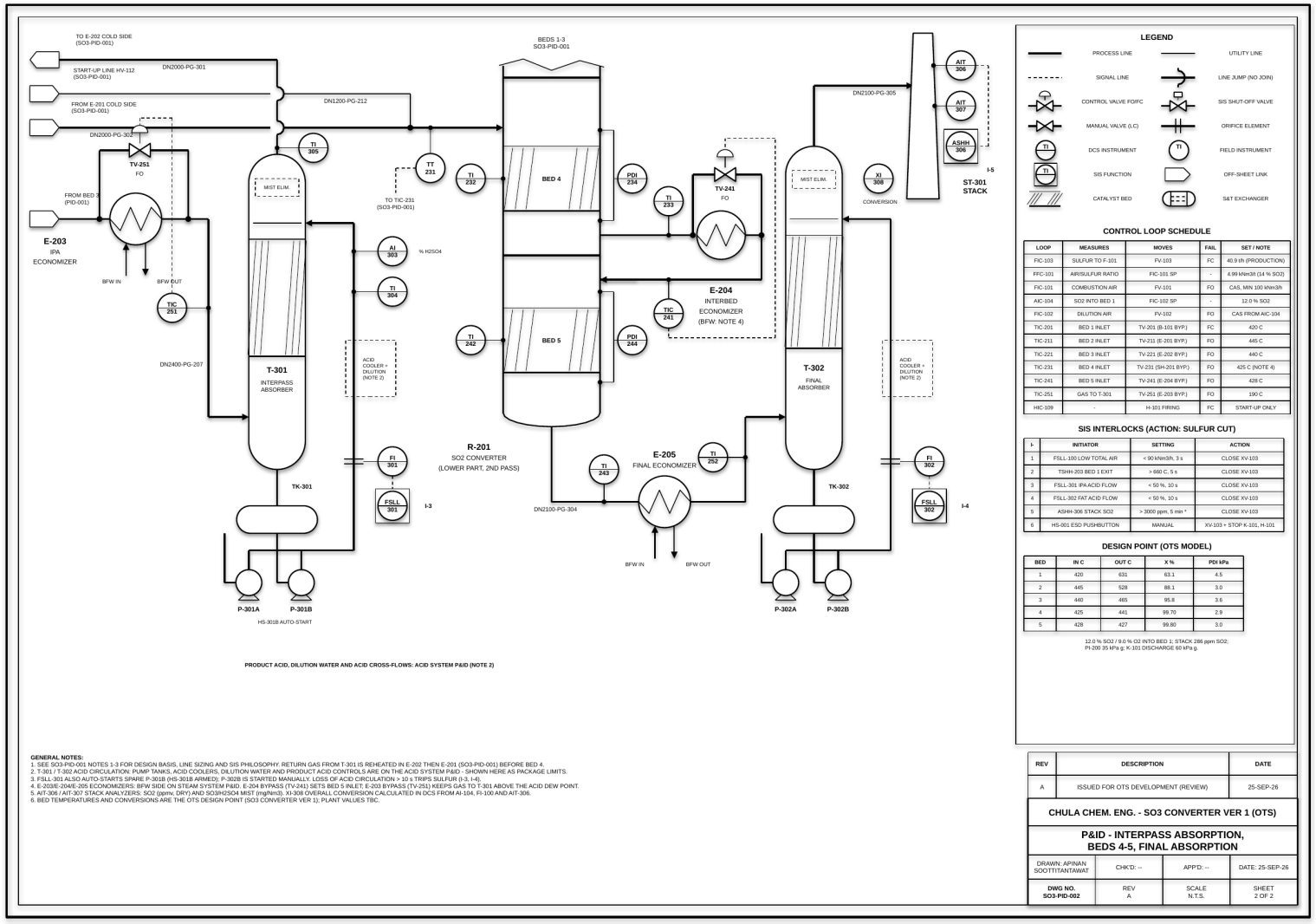
Drawing A2.2 — Screen P&ID of the simulator (element #pidSvg, captured at 2×), the drawing the HAZOP team worked from.

A2.3 EPC P&ID sheet SO3-PID-001 rev A



Drawing A2.3 — EPC P&ID sheet 1 of 2 (SO3-PID-001 rev A).

A2.4 EPC P&ID sheet SO3-PID-002 rev A



Drawing A2.4 — EPC P&ID sheet 2 of 2 (SO3-PID-002 rev A).

Appendix 3 Action close-out sheets

One sheet per recommendation. The action party writes the response, ticks the status and signs; the HAZOP leader verifies the close-out.

Action close-out sheet · R01

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R01
Node	N1 Molten sulfur supply	**Action party**	Instrument / Operations
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	Medium 6

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N1.1	Flow NO: No sulfur to the burner	• Running pump P-101A trips (motor, power) • spare P-101B fails to start • XV-103 closed by a spurious SIS trip • FV-103 closes on instrument-air loss (fail closed) • sulfur frozen in the line	Furnace and converter cool; bed 1 falls toward its strike temperature; if not restored, a preheat with H-101 is needed; production loss	• FIC-103 alarm LO 35 t/h • AIC-104 alarm LO 11 % SO_2 • spare P-101B (manual start) • hot standby procedure	**Medium 6**
N1.7	Pressure LESS: Pump suction pressure low	• Low sulfur pit level • suction strainer plugged • suction line partly frozen	Cavitation, pump damage, loss of flow (see N1.1)	• FIC-103 alarm LO 35 t/h • pit level (outside the battery limit)	**Medium 6**

Recommendation

Auto-start of the spare sulfur pump P-101B on a stop of P-101A or low FI-103, completing before the furnace and bed 1 cool; test the spare monthly.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R02

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R02
Node	N1 Molten sulfur supply, N9 Utilities	**Action party**	Process / Operations
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	Medium 9

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N1.2	Flow LESS: Less sulfur than set	• Sulfur filter blinding • pump wear • FV-103 sticking • sulfur too viscous (cold or overheated) • low pit level	Lower SO_2 strength and production; the beds cool slowly because FFC-101 follows the measured sulfur	• FIC-103 alarm LO 35 t/h • FIC-103 opens FV-103 • PDG-103 local ΔP gauge on the filter • FFC-101 ratio 4.994 kNm^3/t sets the FIC-101 set point (floor 100 kNm^3/h)	**Medium 6**
N1.4	Temperature LESS: Sulfur too cold	• Steam jacket or tracing failure • low jacket steam pressure • long stop with the line full	Sulfur freezes in the line, filter and burner gun: flow loss and plugging; a line can split when it is re-melted (expansion), releasing molten sulfur: burns, fire, SO_2	• Steam-jacketed line DN80-MS-105 • operator rounds (no temperature indication in the DCS)	**Medium 9**
N1.5	Temperature MORE: Sulfur overheated (above about 160 °C)	Jacket steam at too high a pressure	Viscosity rises steeply: flow falls, FV-103 opens wide; more H_2S released from the sulfur (pit area, outside the battery limit)	• Jacket steam pressure set at the source (outside the battery limit, assumed) • FIC-103 alarm LO 35 t/h	**Low 4**
N9.4	Utility NO: Jacket steam lost	• Steam header fault • trap failures	Sulfur freezes in the jacketed line (see N1.4)	Operator rounds	**Medium 6**

Recommendation

Sulfur temperature: indication with low and high alarms on the sulfur line at the pump discharge and at the burner (sulfur solidifies near 115–120 °C and becomes viscous above about 160 °C, assumed values); steam-trap and jacket survey on the round sheet.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R03

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R03
Node	N1 Molten sulfur supply	**Action party**	Instrument / Mechanical
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	Medium 6

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N1.2	Flow LESS: Less sulfur than set	• Sulfur filter blinding • pump wear • FV-103 sticking • sulfur too viscous (cold or overheated) • low pit level	Lower SO_2 strength and production; the beds cool slowly because FFC-101 follows the measured sulfur	• FIC-103 alarm LO 35 t/h • FIC-103 opens FV-103 • PDG-103 local ΔP gauge on the filter • FFC-101 ratio 4.994 kNm^3/t sets the FIC-101 set point (floor 100 kNm^3/h)	**Medium 6**
N1.11	Composition AS WELL AS: Ash, carbon or water in the sulfur	• Poor sulfur quality • filter by-passed or element torn	Ash and carbon carried to the furnace and the first catalyst bed: burner and converter fouling, higher pressure drop over time; carbon burns and uses O_2; water forms acid on cold surfaces	• Sulfur filter • sulfur purchase specification (assumed) • PDI-204 alarm HI 8 kPa	**Medium 6**

Recommendation

Sulfur filter: ΔP transmitter to the DCS with a high alarm (today the local gauge PDG-103 only); duplex filters with a preheated spare.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R04

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R04
Node	N1 Molten sulfur supply, N2 Air supply K-101 and air split FV-101 (combustion) / FV-102 (dilution)	**Action party**	Instrument
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	Medium 9

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N1.3	Flow MORE: More sulfur than intended	• FIC-103 set point raised beyond design (production push) • FV-103 sticks open • FI-103 reads low	More SO_2 and heat; if the air does not follow (ratio station in MAN, FIC-101 at its limit) sulfur vapour leaves the furnace unburned and burns on bed 1: bed-1 overheating, catalyst sintering, stack SO_2 up	• FFC-101 ratio 4.994 kNm^3/t sets the FIC-101 set point (floor 100 kNm^3/h) • FIC-103 alarm HI 44 t/h • TI-203 alarm HI 640 °C • I-2 TSHH-203 trip at bed-1 exit > 660 °C, 5 s (same element as TI-203) • AIT-306 alarm HI 500 ppm	**Medium 9**
N2.4	Flow LESS: Combustion air low against the sulfur	• FIC-101 in MAN at a low output • FI-101 reading high • FFC-101 ratio set low	O_2 in the furnace falls; unburned sulfur vapour carried forward burns on bed 1 or deposits in B-101; lower conversion	• FFC-101 ratio 4.994 kNm^3/t sets the FIC-101 set point (floor 100 kNm^3/h) • AI-105 alarm LO 8 % O_2 • AIC-104 alarm HI 12.8 % SO_2 • AIT-306 alarm HI 500 ppm	**Medium 9**

Recommendation

Cross-limiting of sulfur and combustion air: sulfur demand limited by the measured air (air leads on increase, sulfur leads on decrease), in addition to the FFC-101 ratio station.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R05

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R05
Node	N1 Molten sulfur supply	**Action party**	Instrument
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	Medium 9

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N1.3	Flow MORE: More sulfur than intended	• FIC-103 set point raised beyond design (production push) • FV-103 sticks open • FI-103 reads low	More SO_2 and heat; if the air does not follow (ratio station in MAN, FIC-101 at its limit) sulfur vapour leaves the furnace unburned and burns on bed 1: bed-1 overheating, catalyst sintering, stack SO_2 up	• FFC-101 ratio 4.994 kNm^3/t sets the FIC-101 set point (floor 100 kNm^3/h) • FIC-103 alarm HI 44 t/h • TI-203 alarm HI 640 °C • I-2 TSHH-203 trip at bed-1 exit > 660 °C, 5 s (same element as TI-203) • AIT-306 alarm HI 500 ppm	**Medium 9**

Recommendation

Verify the sulfur flow meter FI-103 (mass meter, proving, comparison with the pit inventory); the FFC-101 ratio and the production figure both depend on it.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R06

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R06
Node	N1 Molten sulfur supply	**Action party**	Process / Instrument
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	Medium 9

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N1.6	Pressure MORE: Pump discharge pressure high (dead-head)	• XV-103 or FV-103 closed with the pump running • filter blocked • burner gun plugged	Pump overheats at no flow; seal failure; molten sulfur leak at the pump: burns, sulfur fire, SO_2	• Pump shut-off head below the line design pressure (assumed) • operator • P-101A/B stop from the DCS	**Medium 9**
N1.7	Pressure LESS: Pump suction pressure low	• Low sulfur pit level • suction strainer plugged • suction line partly frozen	Cavitation, pump damage, loss of flow (see N1.1)	• FIC-103 alarm LO 35 t/h • pit level (outside the battery limit)	**Medium 6**
N1.8	Flow REVERSE: Hot furnace gas back into the sulfur line	• Both pumps stopped with XV-103 open • discharge check valve passing	Hot SO_2 gas flows back toward the pumps and the pit: toxic release in the sulfur area, sulfur fire in the line	• XV-103 closes on every SIS trip • FV-103 fails closed • check valves on the pump discharge (assumed)	**Medium 8**

Recommendation

Sulfur pump and line protection: minimum-flow recycle to the pit; confirm discharge check valves; stop the running pump when XV-103 has been closed for a set time; close XV-103 when no sulfur pump runs.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R07

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R07
Node	N1 Molten sulfur supply	**Action party**	Operations
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	Medium 9

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N1.9	Integrity AS WELL AS: Molten sulfur leak	• Pump seal failure • steam-jacket leak • filter opened while hot or under pressure during a change-over	Molten sulfur at 140 °C: burns; sulfur fire producing SO_2; possible H_2S	• XV-103 remote isolation (SIS) • pump stop from the DCS • I-6 HS ESD push-button (XV-103 closed, K-101 and H-101 stopped) • PPE	**Medium 9**
N1.10	Maintenance OTHER THAN: Filter or pump change-over done incorrectly	• Spare filter not preheated (sulfur freezes) • valve line-up wrong • running pump stopped before the spare runs	Flow interruption (see N1.1); sulfur spill at the filter	• Procedure • FIC-103 alarm LO 35 t/h	**Medium 6**

Recommendation

Written change-over procedure for sulfur filters and pumps (preheated spare, drain, permit); PPE and steam lances at the sulfur area.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R08

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R08
Node	N2 Air supply K-101 and air split FV-101 (combustion) / FV-102 (dilution)	**Action party**	Instrument
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	High 12

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N2.1	Flow NO: No air (blower stops)	• K-101 motor trip (power dip, motor protection) • blower suction blocked (drying tower, outside the battery limit)	Sulfur keeps flowing into a hot furnace without air: sulfur vapour accumulates in F-101, B-101 and the duct and can ignite violently when air returns; no gas through the beds	• I-1 FSLL-100 trip (XV-103) at total air < 90 kNm^3/h, 3 s • XV-103 closes • FI-100 alarm LO 200 kNm^3/h • I-6 HS ESD push-button (XV-103 closed, K-101 and H-101 stopped)	**High 12**

Recommendation

Diversity for I-1: add the K-101 motor-running contact as a second input (1oo2 with FSLL-100) and give FSLL-100 its own flow element, separate from the FI-100 indication.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R09

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R09
Node	N2 Air supply K-101 and air split FV-101 (combustion) / FV-102 (dilution)	**Action party**	Process
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	Medium 8

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N2.2	Flow LESS: Total air low	• Blower degradation • hot afternoon (less mass at the same valve openings) • higher system resistance (bed pressure drop, drying tower)	O_2 short in the furnace and the converter: conversion falls, stack SO_2 rises; at the limit FSLL-100 trips	• FI-100 alarm LO 200 kNm^3/h • AI-105 alarm LO 8 % O_2 • AIT-306 alarm HI 500 ppm • I-1 FSLL-100 trip (XV-103) at total air < 90 kNm^3/h, 3 s	**Medium 8**
N2.8	Temperature MORE: Air at the blower hotter than design	• Hot afternoon • drying-tower acid hot (outside the battery limit)	Less mass at the same volume: lower O_2 and a slightly hotter bed-1 inlet	• FIC-101 and FIC-102 flow control • AI-105 alarm LO 8 % O_2	**Low 4**

Recommendation

Check the capacity of K-101 and of FV-101/FV-102 at the hottest ambient and at end-of-run bed pressure drops.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R10

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R10
Node	N2 Air supply K-101 and air split FV-101 (combustion) / FV-102 (dilution), N4 Waste heat boiler B-101, hot-gas bypass TV-201 and dilution to bed 1	**Action party**	Process / Instrument
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	Medium 9

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N2.4	Flow LESS: Combustion air low against the sulfur	• FIC-101 in MAN at a low output • FI-101 reading high • FFC-101 ratio set low	O_2 in the furnace falls; unburned sulfur vapour carried forward burns on bed 1 or deposits in B-101; lower conversion	• FFC-101 ratio 4.994 kNm^3/t sets the FIC-101 set point (floor 100 kNm^3/h) • AI-105 alarm LO 8 % O_2 • AIC-104 alarm HI 12.8 % SO_2 • AIT-306 alarm HI 500 ppm	**Medium 9**
N2.5	Flow LESS: Dilution air low or lost	• FIC-102 in MAN at a low output • AIC-104 driving FV-102 closed on a wrong analyser value • FV-102 actuator fault	SO_2 into bed 1 above 12 % with O_2 below 9 %: the O_2/SO_2 ratio limits conversion, stack SO_2 rises several times above the permit (see D6)	• AIC-104 alarm HI 12.8 % SO_2 • AI-105 alarm LO 8 % O_2 • AIT-306 alarm HI 500 ppm • I-5 ASHH-306 trip at stack SO_2 > 3000 ppm for 5 min (inhibited 60 min after admission)	**Medium 9**
N3.6	Composition PART OF: Incomplete sulfur combustion	• O_2 short (see N2.4) • poor atomisation at the burner gun	Sulfur deposits in B-101 and on the bed-1 screen; burning on bed 1 gives a hot spot	• AI-105 alarm LO 8 % O_2 • TI-203 alarm HI 640 °C • I-2 TSHH-203 trip at bed-1 exit > 660 °C, 5 s (same element as TI-203)	**Medium 6**
N4.6	Composition MORE: SO_2 strength into bed 1 high	• Dilution air low (see N2.5) • SO_2 analyser error • sulfur surge	Larger bed-1 temperature rise; O_2/SO_2 ratio low; conversion falls and the stack SO_2 rises	• AIC-104 alarm HI 12.8 % SO_2 • AI-105 alarm LO 8 % O_2 • TI-203 alarm HI 640 °C • I-2 TSHH-203 trip at bed-1 exit > 660 °C, 5 s (same element as TI-203) • I-5 ASHH-306 trip at stack SO_2 > 3000 ppm for 5 min (inhibited 60 min after admission)	**Medium 9**
N4.7	Composition LESS: O_2 into bed 1 low	• Combustion air low • SO_2 strength high	Conversion limited by O_2; stack SO_2 above the permit	• AI-105 alarm LO 8 % O_2 • AIT-306 alarm HI 500 ppm	**Medium 6**

Recommendation

Low-low O_2 alarm below the AI-105 low alarm with a written response; evaluate a low-O_2 action on the sulfur by LOPA.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R11

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R11
Node	N2 Air supply K-101 and air split FV-101 (combustion) / FV-102 (dilution)	**Action party**	Instrument / Operations
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	Medium 6

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N2.6	Flow MORE: Dilution air high	• FV-102 fails open (instrument-air loss) • AIC-104 set point lowered • analyser reading high	Weaker gas and a colder bed-1 inlet: less heat for the beds, the second pass can fall toward its strike temperature; production loss	• AIC-104 alarm LO 11 % SO_2 • TIC-201 alarm LO 405 / HI 435 °C • TIC-231 alarm LO 405 / HI 445 °C	**Medium 6**

Recommendation

Engineering limits on the AIC-104 set point and on the FIC-102 remote set point; the SO_2 strength is changed only by written instruction.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R12

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R12
Node	N2 Air supply K-101 and air split FV-101 (combustion) / FV-102 (dilution)	**Action party**	Mechanical / Electrical
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	Medium 8

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N2.3	Flow MORE: Total air high	• FV-101 and FV-102 fail open (instrument-air loss) • wrong FIC-101 set point or ratio entered	Furnace cooler, SO_2 strength lower, more cold air into bed 1: the beds can fall toward their strike temperature; blower motor overload	• FI-100 alarm HI 265 kNm^3/h • AIC-104 alarm LO 11 % SO_2 • TIC-201 alarm LO 405 / HI 435 °C	**Medium 6**
N2.7	Pressure MORE: Blower discharge pressure high (surge)	• FV-101 and FV-102 closing together • high bed pressure drop • blocked duct	Blower surge, vibration and damage; duct overpressure, SO_2 leaks at flanges downstream	• FV-101/FV-102 fail open • PI-200 alarm HI 42 kPa g • PI-101 indication	**Medium 6**
N2.10	Flow REVERSE: Hot SO_2 gas back through the blower	• K-101 stops while the converter system is under pressure • no non-return damper	Hot SO_2-laden gas flows back to the blower and the drying tower: toxic exposure in the blower house	I-1 FSLL-100 trip (XV-103) at total air < 90 kNm^3/h, 3 s stops the sulfur, so no new SO_2 is made	**Medium 8**

Recommendation

K-101 protection: anti-surge or minimum-flow protection, PI-101 high alarm, motor current alarm, and a non-return damper on the discharge that closes when the blower stops.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R13

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R13
Node	N2 Air supply K-101 and air split FV-101 (combustion) / FV-102 (dilution)	**Action party**	Process
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	Medium 9

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N2.9	Composition AS WELL AS: Moisture in the air	• Drying-tower upset (weak drying acid, low circulation • outside the battery limit)	Water forms sulfuric-acid mist with SO_3: corrosion of ducts, exchangers and the stack, acid-mist emission	• AIT-307 alarm HI 40 mg/Nm^3 • drying-tower instruments (outside the battery limit)	**Medium 9**

Recommendation

Moisture analyser on the dried air at the battery limit with a high alarm.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R14

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R14
Node	N3 Sulfur furnace F-101 with preheat burner H-101	**Action party**	Process / Instrument
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	High 12

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N3.1	Temperature MORE: Furnace too hot	• Air/sulfur ratio low (less excess air) • production push • H-101 left firing with sulfur in	Refractory damage, shell hot spots; hotter gas to B-101: tube-sheet overheating	• TI-106 indication (no alarm in the alarm list) • FFC-101 ratio 4.994 kNm^3/t sets the FIC-101 set point (floor 100 kNm^3/h) • TI-107 alarm HI 340 °C	**Medium 6**
N3.2	Temperature LESS: Furnace below sulfur ignition	• Sulfur admitted into a furnace that is not hot enough • a cold furnace after a long stop • large excess air	Sulfur does not ignite and pools in the furnace; the pool later ignites at once: temperature surge, sulfur vapour carry-over, SO_2 puff, overpressure	• start permissive TI-106 ≥ 450 °C • start permissive FIC-103 in MAN ≤ 25 % • nothing acts once sulfur is in	**High 12**
N3.7	Pressure MORE: Furnace pressure surge	• Delayed ignition of a sulfur pool • blocked downstream	Refractory and duct damage; SO_2 leaks at flanges and sight ports	PI-200 alarm HI 42 kPa g	**Medium 6**
N3.9	Measurement OTHER THAN: TI-106 reads wrong	Thermocouple burned out or drifted (single element)	Start permissive based on a wrong value: sulfur admitted into a furnace that is too cold, or a hot furnace not seen	None (single element)	**Medium 6**

Recommendation

Furnace temperature: 2oo3 thermocouples for TI-106 with a high alarm, and a low-temperature action that closes XV-103 when the furnace is below sulfur ignition while sulfur is admitted (evaluate by LOPA).

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R15

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R15
Node	N3 Sulfur furnace F-101 with preheat burner H-101	**Action party**	Instrument
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	High 10

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N3.3	Flame NO: H-101 flame failure during preheat	• Fuel supply lost • flame-out at low air • burner fault	Unburned fuel enters the hot furnace; ignition of the accumulated fuel: explosion in F-101 or B-101	• H-101 own flame protection (fuel shut off) • H-101 trips itself when total air < 60 kNm^3/h • purge before relight	**High 10**

Recommendation

H-101 burner management to NFPA 85/86: purge credit before every light-off, flame-scanner proof test, fuel shut-off valve leak test.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R16

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R16
Node	N3 Sulfur furnace F-101 with preheat burner H-101, N8 Economizer E-205, final absorber T-302, stack ST-301 and stack analysers	**Action party**	Safety
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	Medium 9

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N1.9	Integrity AS WELL AS: Molten sulfur leak	• Pump seal failure • steam-jacket leak • filter opened while hot or under pressure during a change-over	Molten sulfur at 140 °C: burns; sulfur fire producing SO_2; possible H_2S	• XV-103 remote isolation (SIS) • pump stop from the DCS • I-6 HS ESD push-button (XV-103 closed, K-101 and H-101 stopped) • PPE	**Medium 9**
N3.7	Pressure MORE: Furnace pressure surge	• Delayed ignition of a sulfur pool • blocked downstream	Refractory and duct damage; SO_2 leaks at flanges and sight ports	PI-200 alarm HI 42 kPa g	**Medium 6**
N3.8	Integrity AS WELL AS: SO_2 leak from the furnace shell	• Refractory failure (hot spot) • corrosion where the shell is below the acid dew point	SO_2 release in the unit: toxic exposure of people nearby	• Shell temperature survey (assumed) • no fixed SO_2 detection on the P&ID	**Medium 8**

Recommendation

Fixed SO_2 gas detectors at the furnace front, the sulfur pumps, B-101, the converter manways and the acid pumps, alarmed in the control room.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R17

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R17
Node	N4 Waste heat boiler B-101, hot-gas bypass TV-201 and dilution to bed 1, N9 Utilities	**Action party**	Process / Instrument
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	Medium 9

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N4.3	Flow NO: No water on the boiler side	BFW pump failure or drum level control failure (outside the battery limit)	Dry fire tubes with 1121 °C gas: tube and tube-sheet failure; steam explosion risk when water returns; water and steam into the gas: acid corrosion downstream	• Steam-drum level protection (outside the battery limit, assumed) • TI-107 alarm HI 340 °C	**Medium 8**
N6.6	Temperature MORE: Gas to T-301 too hot	• TV-251 fails open (instrument-air loss) • economizer BFW lost • E-203 fouled	Hot gas into the tower: lining and packing damage, acid evaporation and mist	• TIC-251 alarm LO 165 / HI 230 °C • FI-253 alarm LO 110 t/h	**Medium 6**
N9.3	Utility LESS: Economizer BFW low	• BFW pump trip • control fault (outside the battery limit)	E-203 / E-204 / E-205 gas outlets hotter; steaming in the economizers (hammer, tube damage); hotter gas into the IPA and the FAT	• FI-253 alarm LO 110 t/h • TIC-251 alarm HI 230 °C	**Medium 9**
N9.5	Utility MORE: Steam header pressure high	Steam user trips (outside the battery limit)	Steam-drum safety valves lift; B-101 at higher pressure and temperature	Steam-drum safety valves (outside the battery limit, assumed)	**Low 4**

Recommendation

Boiler and economizer water side: confirm that the steam-drum low-low level trip (outside the battery limit) closes XV-103 and add it to the C&E matrix; low-low economizer BFW action and steaming protection.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R18

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R18
Node	N4 Waste heat boiler B-101, hot-gas bypass TV-201 and dilution to bed 1, N5 Converter first pass, N6 Economizer E-203, intermediate absorber T-301 and IPA acid circulation	**Action party**	Mechanical / Process
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	Medium 9

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N4.4	Integrity AS WELL AS: B-101 tube leak (water into the gas)	• Tube corrosion • thermal fatigue at the tube sheet	Steam in the process gas forms acid with SO_3 downstream: corrosion of exchangers and towers, weak acid, catalyst wetting	• Steam/BFW balance (assumed) • AIT-307 alarm HI 40 mg/Nm^3	**Medium 6**
N4.9	Heat transfer LESS: B-101 removes less heat	• Gas-side fouling • low drum level	TV-201 closes to compensate; once it is closed the bed-1 inlet rises with no control left	• TI-107 alarm HI 340 °C • TIC-201 alarm LO 405 / HI 435 °C	**Medium 6**
N5.7	Flow PART OF: Gas bypasses catalyst inside the converter	• Division-plate or bed-seal failure • settled catalyst leaving a gap at the wall	Part of the gas is not converted; overall conversion falls; stack SO_2 above the permit	• AIT-306 alarm HI 500 ppm • XI-308 calculated conversion • I-5 ASHH-306 trip at stack SO_2 > 3000 ppm for 5 min (inhibited 60 min after admission)	**Medium 6**
N5.9	Integrity AS WELL AS: E-201 or E-202 tube leak	• Acid corrosion at the cold end • thermal fatigue • tube-to-tubesheet weld failure	First-pass gas leaks into the return gas and skips the IPA: overall conversion falls, stack SO_2 and acid mist rise; corrosion	• AIT-306 alarm HI 500 ppm • I-5 ASHH-306 trip at stack SO_2 > 3000 ppm for 5 min (inhibited 60 min after admission)	**Medium 9**
N6.9	Integrity AS WELL AS: E-203 tube leak (BFW into the gas)	• Cold-end corrosion • thermal fatigue	Water into SO_3 gas: acid in the duct and the IPA; corrosion; weak acid	• AI-303 alarm LO 98 % (late) • FI-253 against the steam balance	**Medium 6**
N7.3	Composition AS WELL AS: Unconverted gas reaches the second pass	• Start-up line HV-112 left open or passing • interpass exchanger tube leak (see N5.9) • internal bypass in the first pass (see N5.7)	Final conversion falls; stack SO_2 rises above the 500 ppm permit for as long as it lasts	• HV-112 locked closed in operation • start permissive HV-112 closed • AIT-306 alarm HI 500 ppm • I-5 ASHH-306 trip at stack SO_2 > 3000 ppm for 5 min (inhibited 60 min after admission)	**Medium 9**

Recommendation

Tube-leak detection and turnaround inspection: steam/BFW balance and conductivity, leak tests of B-101 and E-201 to E-205, inspection of the converter division plates and bed seals.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R19

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R19
Node	N4 Waste heat boiler B-101, hot-gas bypass TV-201 and dilution to bed 1	**Action party**	Instrument
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	Medium 9

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N4.6	Composition MORE: SO_2 strength into bed 1 high	• Dilution air low (see N2.5) • SO_2 analyser error • sulfur surge	Larger bed-1 temperature rise; O_2/SO_2 ratio low; conversion falls and the stack SO_2 rises	• AIC-104 alarm HI 12.8 % SO_2 • AI-105 alarm LO 8 % O_2 • TI-203 alarm HI 640 °C • I-2 TSHH-203 trip at bed-1 exit > 660 °C, 5 s (same element as TI-203) • I-5 ASHH-306 trip at stack SO_2 > 3000 ppm for 5 min (inhibited 60 min after admission)	**Medium 9**
N4.8	Measurement OTHER THAN: SO_2 analyser AI-104 wrong	• Calibration drift • sample line blocked with sulfur or acid • analyser power loss	AIC-104 moves FV-102 the wrong way: the real gas strength is away from 12 % (see N4.6 and N2.6)	• AIC-104 alarm LO 11 / HI 12.8 % SO_2 • AI-105 alarm LO 8 % O_2 • FIC-102 range 0–80 kNm^3/h	**Medium 9**

Recommendation

Analyser validation for AI-104 and AI-105: calibration schedule, sample-system checks and comparison with the SO_2 strength calculated from FI-103 and FI-100.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R20

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R20
Node	N5 Converter first pass	**Action party**	Instrument / Safety
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	High 12

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N4.2	Temperature LESS: Bed-1 inlet too cold	• TV-201 fails closed (instrument-air loss) • TIC-201 measurement error • dilution air high • furnace cold	Bed 1 approaches its strike temperature (≈375 °C): conversion collapses, unconverted SO_2 to the stack	• TIC-201 alarm LO 405 / HI 435 °C • AIT-306 alarm HI 500 ppm • I-5 ASHH-306 trip at stack SO_2 > 3000 ppm for 5 min (inhibited 60 min after admission)	**Medium 9**
N5.1	Temperature MORE: Bed-1 exit above the catalyst limit	• Bed-1 inlet too hot (see N4.1) • SO_2 strength high • unburned sulfur burning on bed 1	Catalyst sintering and permanent activity loss; active-phase migration; grids and division plates distort; shell overstress	• TI-203 alarm HI 640 °C • I-2 TSHH-203 trip at bed-1 exit > 660 °C, 5 s (same element as TI-203) • TIC-201 alarm LO 405 / HI 435 °C	**Medium 9**
N5.2	Measurement OTHER THAN: TI-203 element fails low	• Open-circuit thermocouple • wiring fault	The TI-203 alarm and the TSHH-203 trip read the same element, so both are blind: a real bed-1 over-temperature goes undetected (see D5: bed-1 middle near 790 °C, no TSHH-203 trip)	• TI-202 bed-1 middle indication (no alarm) • AIT-306 alarm HI 500 ppm late	**High 12**

Recommendation

TSHH-203 on dedicated elements (2oo3), separate from the TI-203 indication and alarm, with bad-PV and deviation alarms between the bed thermocouples and a TI-202 high alarm; LOPA for bed-1 over-temperature.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R21

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R21
Node	N4 Waste heat boiler B-101, hot-gas bypass TV-201 and dilution to bed 1, N5 Converter first pass, N7 Return gas (E-202 / E-201 cold sides), beds 4–5 and E-204	**Action party**	Process
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	Medium 9

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N1.11	Composition AS WELL AS: Ash, carbon or water in the sulfur	• Poor sulfur quality • filter by-passed or element torn	Ash and carbon carried to the furnace and the first catalyst bed: burner and converter fouling, higher pressure drop over time; carbon burns and uses O_2; water forms acid on cold surfaces	• Sulfur filter • sulfur purchase specification (assumed) • PDI-204 alarm HI 8 kPa	**Medium 6**
N4.5	Pressure MORE: Converter inlet pressure high	• Rising bed pressure drop (dust, catalyst fines) • blockage downstream • mist-eliminator plugging	The blower moves back on its curve: less air; duct and converter overpressure, leaks	• PI-200 alarm HI 42 kPa g • PDI-204 alarm HI 8 kPa • PI-101 indication	**Medium 6**
N5.6	Pressure MORE: Bed pressure drop high	• Dust and ash on the bed screens • catalyst breakage from thermal cycling	Higher PI-200, less air from K-101; maldistribution and channelling give hot spots	• PDI-204 alarm HI 8 kPa • PDI-214, PDI-224 indications • PI-200 alarm HI 42 kPa g	**Medium 6**
N5.8	Catalyst activity LESS: Catalyst activity low in beds 1–3	• Ageing • dust • poisoning (arsenic, chlorides, water) • past overheating	Less conversion per bed, more work left to the later beds; stack SO_2 up; early catalyst replacement	• Bed temperature indications • AIT-306 alarm HI 500 ppm	**Medium 6**
N7.6	Catalyst activity LESS: Bed-4 catalyst activity low	• Catalyst wetted by acid during a shutdown • ageing • dust	Stack SO_2 rises toward and above the permit but stays below the ASHH-306 trip (see D9): a long exceedance with no trip	• AIT-306 alarm HI 500 ppm • I-5 ASHH-306 trip at stack SO_2 > 3000 ppm for 5 min (inhibited 60 min after admission) only far above the permit	**Medium 9**
N7.7	Pressure MORE: Bed 4 / bed 5 pressure drop high	• Dust • sulfate crusts after acid wetting	Higher system pressure and less air	• PDI-234, PDI-244 indications • PI-200 alarm HI 42 kPa g	**Low 3**

Recommendation

Catalyst management: ΔP trend of every bed, PDI-214 to PDI-244 high alarms, screening at turnarounds, activity sampling and a replacement plan.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R22

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R22
Node	N5 Converter first pass, N7 Return gas (E-202 / E-201 cold sides), beds 4–5 and E-204	**Action party**	Instrument
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	Medium 9

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N4.1	Temperature MORE: Bed-1 inlet too hot	• TV-201 fails open or its seat leaks • TIC-201 set point too high • B-101 fouled (gas leaves hotter) • TIC-201 measurement error	Bed-1 exit above the catalyst limit: sintering and permanent activity loss, lower bed-1 conversion, converter internals and shell overheated	• TIC-201 alarm LO 405 / HI 435 °C • TI-203 alarm HI 640 °C • I-2 TSHH-203 trip at bed-1 exit > 660 °C, 5 s (same element as TI-203) • TV-201 fails closed on air loss	**Medium 9**
N4.2	Temperature LESS: Bed-1 inlet too cold	• TV-201 fails closed (instrument-air loss) • TIC-201 measurement error • dilution air high • furnace cold	Bed 1 approaches its strike temperature (≈375 °C): conversion collapses, unconverted SO_2 to the stack	• TIC-201 alarm LO 405 / HI 435 °C • AIT-306 alarm HI 500 ppm • I-5 ASHH-306 trip at stack SO_2 > 3000 ppm for 5 min (inhibited 60 min after admission)	**Medium 9**
N5.3	Temperature LESS: Bed-2 or bed-3 inlet too cold	• TV-211 or TV-221 stuck or closed • TIC-211/TIC-221 set point low • bypass valve sticking	Bed approaches its strike temperature; conversion falls, stack SO_2 rises	• TIC-211 alarm LO 430 / HI 460 °C • TIC-221 alarm LO 425 / HI 455 °C • AIT-306 alarm HI 500 ppm	**Medium 6**
N5.4	Temperature MORE: Bed-2 or bed-3 inlet too hot	• TV-211 / TV-221 fail open (instrument-air loss) • exchanger hot side fouled	Lower equilibrium conversion in beds 2–3; more SO_2 to the IPA and the second pass; stack SO_2 up	• TIC-211 alarm HI 460 °C • TIC-221 alarm HI 455 °C • AIT-306 alarm HI 500 ppm	**Medium 6**
N5.5	Control OTHER THAN: A bed-inlet loop oscillates	• Bypass valve stick-slip • poor tuning • interaction between the bed-inlet loops through the return gas	Bed temperatures and conversion swing; thermal fatigue of exchangers	• TIC-211 alarm LO 430 / HI 460 °C • TIC-221 alarm LO 425 / HI 455 °C • tuning records	**Medium 6**
N7.2	Temperature MORE: Bed-4 inlet too hot	• TV-231 fails open (instrument-air loss), so more heat reaches E-201 • high production	Lower equilibrium conversion in bed 4; stack SO_2 up	• TIC-231 alarm HI 445 °C • AIT-306 alarm HI 500 ppm	**Medium 6**
N7.4	Temperature LESS: Bed-5 inlet too cold	• TV-241 closed or stuck • E-204 removing more heat than needed	Bed 5 toward its strike temperature; final conversion falls	• TIC-241 alarm LO 410 / HI 445 °C • AIT-306 alarm HI 500 ppm	**Medium 6**

Recommendation

Diagnostics on the bypass valves TV-201 to TV-251 (position feedback against controller output, partial stroke at turnaround) and loop-performance monitoring.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			

	Name	Signature	Date
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R23

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R23
Node	N5 Converter first pass, N6 Economizer E-203, intermediate absorber T-301 and IPA acid circulation, N8 Economizer E-205, final absorber T-302, stack ST-301 and stack analysers	**Action party**	Instrument
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	Medium 6

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N5.10	Temperature MORE: Superheated steam too hot	• Low steam flow through SH-201 (less steam from B-101) • TV-231 closed	Superheater tube overheating; damage to the steam user (outside the battery limit)	• TI-207 indication (no alarm) • TIC-231 bypass	**Medium 6**
N6.4	Composition MORE: IPA acid strength high	Dilution water lost	SO_3 slip over strong acid; acid mist	AI-303 indication (no high alarm)	**Low 4**
N6.6	Temperature MORE: Gas to T-301 too hot	• TV-251 fails open (instrument-air loss) • economizer BFW lost • E-203 fouled	Hot gas into the tower: lining and packing damage, acid evaporation and mist	• TIC-251 alarm LO 165 / HI 230 °C • FI-253 alarm LO 110 t/h	**Medium 6**
N6.10	Flow MORE: IPA acid flow high	Both pumps running	Tower flooding; acid carry-over to the E-202 cold side: corrosion	FI-301 indication (no high alarm)	**Low 4**
N8.5	Temperature MORE: Gas to T-302 too hot	• Economizer BFW lost • E-205 fouled	Poorer final absorption; tower lining damage	• TI-252 indication (no alarm) • FI-253 alarm LO 110 t/h	**Medium 6**

Recommendation

Add DCS alarms where today there is only an indication: TI-207 high, TI-305 high, TI-252 high, AI-303 high, FI-301 high.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R24

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R24
Node	N6 Economizer E-203, intermediate absorber T-301 and IPA acid circulation, N8 Economizer E-205, final absorber T-302, stack ST-301 and stack analysers	**Action party**	Instrument / Operations
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	High 12

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N6.1	Flow NO: No IPA acid circulation	• P-301A trips and the P-301B auto-start fails • suction valve closed • pump tank empty	SO_3 passes the tower unabsorbed: dense acid mist from the stack and SO_3 into the second pass; hot dry gas damages the tower lining; stack SO_2 up (see D7, D8)	• HS-301B auto-start of P-301B • FI-301 alarm LO 1200 m^3/h • I-3 FSLL-301 trip at IPA acid < 800 m^3/h, 10 s	**Medium 8**
N8.1	Flow NO: No FAT acid circulation	• P-302A trips (P-302B is a manual spare) • suction blocked	SO_3 goes up the stack as a dense acid-mist plume that can reach people downwind (see D8: AIT-307 far above its alarm before the trip); tower lining damage	• FI-302 alarm LO 750 m^3/h • I-4 FSLL-302 trip at FAT acid < 500 m^3/h, 10 s • AIT-307 alarm HI 40 mg/Nm^3	**High 12**
N8.2	Flow LESS: FAT acid flow low	• Pump wear • strainer or distributor plugging	Poorer final absorption; acid mist	• FI-302 alarm LO 750 m^3/h • I-4 FSLL-302 trip at FAT acid < 500 m^3/h, 10 s	**Medium 6**

Recommendation

Acid pump spares: monthly proof test of the HS-301B auto-start, and an auto-start for P-302B on FSLL-302 like HS-301B.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R25

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R25
Node	N6 Economizer E-203, intermediate absorber T-301 and IPA acid circulation	**Action party**	Instrument
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	Medium 8

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N6.1	Flow NO: No IPA acid circulation	• P-301A trips and the P-301B auto-start fails • suction valve closed • pump tank empty	SO_3 passes the tower unabsorbed: dense acid mist from the stack and SO_3 into the second pass; hot dry gas damages the tower lining; stack SO_2 up (see D7, D8)	• HS-301B auto-start of P-301B • FI-301 alarm LO 1200 m^3/h • I-3 FSLL-301 trip at IPA acid < 800 m^3/h, 10 s	**Medium 8**
N6.2	Flow LESS: IPA acid flow low	• Pump wear • strainer or distributor plugging	Poorer absorption, acid mist; dry spots in the packing	• FI-301 alarm LO 1200 m^3/h • I-3 FSLL-301 trip at IPA acid < 800 m^3/h, 10 s	**Medium 6**

Recommendation

Level transmitters with low alarm on the acid pump tanks TK-301 and TK-302.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R26

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R26
Node	N6 Economizer E-203, intermediate absorber T-301 and IPA acid circulation	**Action party**	Process
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	Medium 9

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N6.3	Composition LESS: IPA acid strength low	• Dilution water valve passing • cross-over from another acid system • wrong sample	SO_3 absorption falls: acid mist carried forward and emitted; faster corrosion of acid piping (weaker acid)	• AI-303 alarm LO 98 % • AIT-307 alarm HI 40 mg/Nm^3	**Medium 9**
N6.4	Composition MORE: IPA acid strength high	Dilution water lost	SO_3 slip over strong acid; acid mist	AI-303 indication (no high alarm)	**Low 4**

Recommendation

Dilution water to the IPA acid: double block valves, flow indication and an acid-strength control alarm.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R27

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R27
Node	N6 Economizer E-203, intermediate absorber T-301 and IPA acid circulation, N9 Utilities	**Action party**	Process
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	Medium 6

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N6.5	Temperature MORE: IPA acid too hot	• Acid cooler fouled • cooling water lost (outside the battery limit)	Absorption falls (mist); corrosion of acid piping and coolers rises sharply	TI-304 alarm HI 95 °C	**Medium 6**
N9.6	Utility NO: Cooling water to the acid coolers lost	Cooling-water pump trip (outside the battery limit)	Acid temperature rises (see N6.5): absorption falls, corrosion rises	TI-304 alarm HI 95 °C	**Medium 6**

Recommendation

Acid-cooler cooling-water flow and ΔT monitoring; anodic-protection alarms.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R28

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R28
Node	N6 Economizer E-203, intermediate absorber T-301 and IPA acid circulation, N8 Economizer E-205, final absorber T-302, stack ST-301 and stack analysers	**Action party**	Process
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	Medium 6

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N6.7	Temperature LESS: Gas to T-301 too cold	• TV-251 closed • low gas flow • TIC-251 set point low	Metal below the acid dew point in E-203: condensation and corrosion; tube leak lets BFW into the gas	TIC-251 alarm LO 165 °C	**Medium 6**
N8.6	Temperature LESS: Gas to T-302 below the acid dew point	Low production rate with full BFW flow (E-205 has no bypass)	Acid condensation in E-205 and the duct: corrosion and tube leaks	None (E-205 has no bypass)	**Medium 6**

Recommendation

Dew-point margin: minimum gas outlet and BFW inlet temperatures for E-203 to E-205, and a gas bypass for E-205 evaluated for low rates.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R29

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R29
Node	N6 Economizer E-203, intermediate absorber T-301 and IPA acid circulation, N8 Economizer E-205, final absorber T-302, stack ST-301 and stack analysers	**Action party**	Safety
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	High 12

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N6.8	Integrity AS WELL AS: Acid leak at the IPA pumps or piping	• Seal failure • corrosion • flange leak	98.5 % acid at 80 °C: severe chemical burns; spill	• I-6 HS ESD push-button (XV-103 closed, K-101 and H-101 stopped) • acid pumps stop from the DCS • PPE, safety showers (assumed)	**High 12**
N8.9	Integrity AS WELL AS: Acid leak at the FAT pumps or piping	• Seal failure • corrosion • flange leak	Concentrated acid: severe chemical burns; spill	• I-6 HS ESD push-button (XV-103 closed, K-101 and H-101 stopped) • pumps stop from the DCS • PPE, safety showers (assumed)	**Medium 8**
N10.9	ESD LATE: Emergency shutdown too late in an acid or SO_2 leak	• Hesitation • no clear criteria	Leak continues: exposure of operators and responders	I-6 HS ESD push-button (XV-103 closed, K-101 and H-101 stopped)	**Medium 8**

Recommendation

Acid leak protection: spray shields on flanges, safety showers and eyewash at the acid pumps, an acid-leak emergency drill on the simulator.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R30

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R30
Node	N7 Return gas (E-202 / E-201 cold sides), beds 4–5 and E-204	**Action party**	Process
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	Medium 8

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N7.1	Temperature LESS: Bed-4 inlet too cold	• Less heat available from beds 1–2 (low rate, weak gas) • TV-231 at its limit • interpass exchangers fouled or bypassing • IPA gas cold	Bed 4 approaches its strike temperature; final conversion falls; stack SO_2 above the permit	• TIC-231 alarm LO 405 / HI 445 °C • AIT-306 alarm HI 500 ppm • I-5 ASHH-306 trip at stack SO_2 > 3000 ppm for 5 min (inhibited 60 min after admission)	**Medium 8**

Recommendation

Operating envelope: minimum sulfur rate and SO_2 strength at which bed 4 holds its inlet temperature without start-up help, written into the operating procedure.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R31

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R31
Node	N7 Return gas (E-202 / E-201 cold sides), beds 4–5 and E-204, N10 Procedures	**Action party**	Operations / Instrument
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	Medium 9

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N7.3	Composition AS WELL AS: Unconverted gas reaches the second pass	• Start-up line HV-112 left open or passing • interpass exchanger tube leak (see N5.9) • internal bypass in the first pass (see N5.7)	Final conversion falls; stack SO_2 rises above the 500 ppm permit for as long as it lasts	• HV-112 locked closed in operation • start permissive HV-112 closed • AIT-306 alarm HI 500 ppm • I-5 ASHH-306 trip at stack SO_2 > 3000 ppm for 5 min (inhibited 60 min after admission)	**Medium 9**
N10.3	Sequence AFTER: HV-112 closed late or not at all	• Step missed • chain not fitted • valve not fully seated	Part of the converter feed skips beds 1–3 and the IPA: final conversion falls, stack SO_2 above the permit	• start permissive HV-112 closed • HV-112 locked closed	**Medium 9**

Recommendation

HV-112: closed-position switch alarmed in the DCS, car-seal register, and the closure signed in the start-up checklist.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R32

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R32
Node	N7 Return gas (E-202 / E-201 cold sides), beds 4–5 and E-204, N8 Economizer E-205, final absorber T-302, stack ST-301 and stack analysers	**Action party**	Mechanical
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	Medium 6

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N7.8	Integrity AS WELL AS: Corrosion of the cold return-gas side	• Acid mist carried over from T-301 (mist eliminator damaged) • metal below the dew point at the E-202 cold end	Duct and E-202 tube corrosion; leaks	• T-301 mist eliminators (assumed) • TI-254 indication	**Medium 6**
N8.4	Composition MORE: Stack acid mist high	• FAT acid strength or temperature off • candle filter failure • moisture in the air	Visible plume, local acid fallout, corrosion of the stack	AIT-307 alarm HI 40 mg/Nm^3	**Medium 6**

Recommendation

Mist eliminators in T-301 and T-302: ΔP indication and inspection of the candles at every turnaround.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R33

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R33
Node	N8 Economizer E-205, final absorber T-302, stack ST-301 and stack analysers	**Action party**	Process / Environment
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	High 12

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N2.5	Flow LESS: Dilution air low or lost	• FIC-102 in MAN at a low output • AIC-104 driving FV-102 closed on a wrong analyser value • FV-102 actuator fault	SO_2 into bed 1 above 12 % with O_2 below 9 %: the O_2/SO_2 ratio limits conversion, stack SO_2 rises several times above the permit (see D6)	• AIC-104 alarm HI 12.8 % SO_2 • AI-105 alarm LO 8 % O_2 • AIT-306 alarm HI 500 ppm • I-5 ASHH-306 trip at stack SO_2 > 3000 ppm for 5 min (inhibited 60 min after admission)	**Medium 9**
N7.6	Catalyst activity LESS: Bed-4 catalyst activity low	• Catalyst wetted by acid during a shutdown • ageing • dust	Stack SO_2 rises toward and above the permit but stays below the ASHH-306 trip (see D9): a long exceedance with no trip	• AIT-306 alarm HI 500 ppm • I-5 ASHH-306 trip at stack SO_2 > 3000 ppm for 5 min (inhibited 60 min after admission) only far above the permit	**Medium 9**
N8.3	Composition MORE: Stack SO_2 above the permit	• Any conversion loss upstream (bed temperatures, catalyst, bypassing, O_2 short) • start-up	Permit exceeded: fines, complaints; at high levels a health risk downwind	• AIT-306 alarm HI 500 ppm • I-5 ASHH-306 trip at stack SO_2 > 3000 ppm for 5 min (inhibited 60 min after admission)	**High 12**

Recommendation

ASHH-306 and the permit: review the set point, delay and start-up inhibit against the toxic consequence downwind (dispersion study), and write a permit-exceedance response procedure.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R34

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R34
Node	N8 Economizer E-205, final absorber T-302, stack ST-301 and stack analysers	**Action party**	Instrument
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	High 12

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N8.3	Composition MORE: Stack SO_2 above the permit	• Any conversion loss upstream (bed temperatures, catalyst, bypassing, O_2 short) • start-up	Permit exceeded: fines, complaints; at high levels a health risk downwind	• AIT-306 alarm HI 500 ppm • I-5 ASHH-306 trip at stack SO_2 > 3000 ppm for 5 min (inhibited 60 min after admission)	**High 12**
N8.7	Measurement OTHER THAN: Stack SO_2 analyser AIT-306 wrong or frozen	• Sample line plugged • calibration drift • range limit (0–5000 ppm)	A permit exceedance is not seen; ASHH-306 reads the same analyser and does not trip	• XI-308 calculated conversion • analyser calibration (assumed)	**Medium 9**
N8.8	Measurement OTHER THAN: Acid-mist analyser AIT-307 wrong	Sample line or optics fouled	A mist emission is not seen	Visual check of the plume	**Medium 6**

Recommendation

Stack analysers: fault alarms for AIT-306 and AIT-307, and an independent SO_2 measurement for ASHH-306 (today the trip and the permit alarm share one analyser).

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R35

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R35
Node	N9 Utilities	**Action party**	Instrument
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	Medium 6

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N9.1	Utility NO: Instrument-air failure	• Compressor trip • dryer failure • header leak	Every valve to its fail position: FV-103 closes (sulfur lost), FV-101/FV-102 open (air to its maximum), TV-201 closes, TV-211 to TV-251 open, H-101 off; no SIS trip, the beds cool fast with full air (see D10)	• Fail-safe valve positions (sulfur off) • FIC-103 alarm LO 35 t/h • FI-100 alarm HI 265 kNm^3/h	**Medium 6**

Recommendation

Instrument air: low-pressure alarm, receiver hold-up time, and confirm that XV-103 fails closed on air loss.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R36

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R36
Node	N9 Utilities	**Action party**	Electrical / Operations
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	Medium 9

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N9.2	Utility NO: Power failure	• Grid loss • switchboard fault	K-101 and all pumps stop; sulfur stops; acid circulation stops with hot gas in the towers; on restart, the blower before the acid pumps blows SO_3 to the stack	• I-1 FSLL-100 trip (XV-103) at total air < 90 kNm^3/h, 3 s • I-3 FSLL-301 trip at IPA acid < 800 m^3/h, 10 s • I-4 FSLL-302 trip at FAT acid < 500 m^3/h, 10 s • SIS de-energise to trip (assumed)	**Medium 9**

Recommendation

Restart after a power failure: acid circulation established before the blower; UPS for DCS and SIS.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R37

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R37
Node	N9 Utilities	**Action party**	Instrument / Safety
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	Medium 8

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N9.7	Safety system NO: XV-103 fails to close on demand	• Valve stuck (sulfur solidified on the seat) • solenoid or actuator fault	Every trip acts on this one valve: the hazard each trip protects against goes ahead	FV-103 closes as well, because the trip puts FIC-103 in MAN at 0 % (control system, not independent)	**Medium 8**

Recommendation

XV-103: proof test with partial stroke; LOPA for a second, diverse sulfur shut-off (1oo2), because every trip acts on this one valve.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R38

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R38
Node	N10 Procedures	**Action party**	Operations
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	Medium 9

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N10.5	Sweep LESS: Shutdown sweep too short	Blower stopped straight after the sulfur stop	SO_2 and SO_3 left in the converter and ducts: released when equipment is opened; SO_3 condenses as acid when the converter cools (catalyst wetting)	Procedure: 5-min sweep at about 100 kNm^3/h with the stack SO_2 falling	**Medium 9**
N10.6	Sweep MORE: Air sweep too long	Blower left running after the sweep	Beds cool below their strike temperatures: a hot restart becomes a cold start-up	Procedure	**Low 4**

Recommendation

Shutdown sweep: end criteria (time and stack SO_2), dry air only, and no prolonged air flow through a cooling converter.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R39

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R39
Node	N3 Sulfur furnace F-101 with preheat burner H-101, N10 Procedures	**Action party**	Operations
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	High 12

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N3.2	Temperature LESS: Furnace below sulfur ignition	• Sulfur admitted into a furnace that is not hot enough • a cold furnace after a long stop • large excess air	Sulfur does not ignite and pools in the furnace; the pool later ignites at once: temperature surge, sulfur vapour carry-over, SO_2 puff, overpressure	• start permissive TI-106 ≥ 450 °C • start permissive FIC-103 in MAN ≤ 25 % • nothing acts once sulfur is in	**High 12**
N3.4	Flow MORE: H-101 firing too high	HIC-109 set high for a fast heat-up	Thermal shock to refractory and catalyst; bed 1 heated above 440 °C before sulfur	• start permissive TIC-201 and TI-202 ≥ 400 °C, inlet ≤ 440 °C • TIC-201 alarm HI 435 °C	**Medium 6**
N3.5	Sequence AS WELL AS: H-101 still firing after sulfur admission	Start-up step missed	Extra heat and combustion water (acid mist, corrosion); O_2 consumed; furnace hotter	• Procedure step • FI-109 indication	**Medium 6**
N10.1	Preheat LESS: Beds not hot enough at sulfur admission	• Permissive bypassed • bed-4 thermocouple wrong • admission before the second pass is hot	SO_2 passes unconverted to the stack during the start-up (ASHH-306 is inhibited); permit exceedance	• start permissive TIC-201 and TI-202 ≥ 400 °C, inlet ≤ 440 °C • start permissive TI-232 ≥ 360 °C • AIT-306 alarm HI 500 ppm	**Medium 6**
N10.2	Sequence BEFORE: Sulfur admitted before acid circulates	Steps taken out of order	SO_3 not absorbed: acid-mist plume	• start permissive FI-301 and FI-302 ≥ 80 % • I-3 FSLL-301 trip at IPA acid < 800 m^3/h, 10 s • I-4 FSLL-302 trip at FAT acid < 500 m^3/h, 10 s	**Medium 6**
N10.4	Sulfur admission MORE: Too much sulfur at admission	FIC-103 not in MAN or output above 25 %	Sulfur slug into a furnace just at ignition: pooling and puff; strong gas on beds that are not lined out	• start permissive FIC-103 in MAN ≤ 25 % • start permissive TI-106 ≥ 450 °C	**Medium 6**

Recommendation

Start-up procedure: maximum heat-up rate, H-101 firing limits, sulfur admission at FIC-103 ≤ 25 % in MAN with a ramp, an alarm when H-101 still fires after sulfur admission, and a start-up emission plan.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R40

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R40
Node	N10 Procedures	**Action party**	Operations
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	High 12

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N6.8	Integrity AS WELL AS: Acid leak at the IPA pumps or piping	• Seal failure • corrosion • flange leak	98.5 % acid at 80 °C: severe chemical burns; spill	• I-6 HS ESD push-button (XV-103 closed, K-101 and H-101 stopped) • acid pumps stop from the DCS • PPE, safety showers (assumed)	**High 12**
N10.8	Trip reset EARLY: SIS reset and sulfur re-admitted before the cause is cleared	• Pressure to restart • first-out not reviewed	Repeat trip; sulfur into a furnace or beds that have cooled; over-temperature if the cause was a hot bed	• Reset blocked while TSHH-203 is active • start permissives at re-admission	**Medium 6**
N10.9	ESD LATE: Emergency shutdown too late in an acid or SO_2 leak	• Hesitation • no clear criteria	Leak continues: exposure of operators and responders	I-6 HS ESD push-button (XV-103 closed, K-101 and H-101 stopped)	**Medium 8**

Recommendation

Emergency procedures: ESD criteria, trip reset only after a first-out review with supervisor authorisation; simulator drills.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R41

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R41
Node	N10 Procedures	**Action party**	Operations / Safety
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	Medium 8

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N10.10	Maintenance OTHER THAN: Equipment opened with SO_2/SO_3 or sulfur present	• Incomplete sweep • isolation incomplete (XV-103 passing, sulfur line not blinded)	Toxic exposure of the maintenance crew; molten sulfur release	Permit to work and gas test (assumed)	**Medium 8**

Recommendation

Isolation and blinding list for the sulfur and gas systems; SO_2 gas test and permit before entry or opening.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R42

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R42
Node	N10 Procedures	**Action party**	Operations
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	High 12

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N10.1	Preheat LESS: Beds not hot enough at sulfur admission	• Permissive bypassed • bed-4 thermocouple wrong • admission before the second pass is hot	SO_2 passes unconverted to the stack during the start-up (ASHH-306 is inhibited); permit exceedance	• start permissive TIC-201 and TI-202 ≥ 400 °C, inlet ≤ 440 °C • start permissive TI-232 ≥ 360 °C • AIT-306 alarm HI 500 ppm	**Medium 6**
N10.11	Safety system AS WELL AS: A trip or permissive left bypassed	Bypass for maintenance or start-up not removed	The protection is missing when the demand comes (over-temperature, loss of air, acid loss)	Administrative control only	**High 12**

Recommendation

Trip and permissive bypass management: permit, time limit, MOC, bypass status shown on the DCS.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			

Action close-out sheet · R43

Project	SO3 Converter Ver 1 · converter section	**Document**	SO3-V1-SA-HZP-001 rev A
Study phase	HAZOP (design / teaching study)	**Action no.**	R43
Node	N4 Waste heat boiler B-101, hot-gas bypass TV-201 and dilution to bed 1, N5 Converter first pass	**Action party**	Process / Instrument
Drawings	SO3-V1-PID-001 rev A; SO3-PID-001 / -002 rev A	**Highest risk addressed**	Medium 9

Raised from

Ref	Deviation	Causes	Consequences	Safeguards	Risk
N4.1	Temperature MORE: Bed-1 inlet too hot	• TV-201 fails open or its seat leaks • TIC-201 set point too high • B-101 fouled (gas leaves hotter) • TIC-201 measurement error	Bed-1 exit above the catalyst limit: sintering and permanent activity loss, lower bed-1 conversion, converter internals and shell overheated	• TIC-201 alarm LO 405 / HI 435 °C • TI-203 alarm HI 640 °C • I-2 TSHH-203 trip at bed-1 exit > 660 °C, 5 s (same element as TI-203) • TV-201 fails closed on air loss	**Medium 9**
N5.1	Temperature MORE: Bed-1 exit above the catalyst limit	• Bed-1 inlet too hot (see N4.1) • SO_2 strength high • unburned sulfur burning on bed 1	Catalyst sintering and permanent activity loss; active-phase migration; grids and division plates distort; shell overstress	• TI-203 alarm HI 640 °C • I-2 TSHH-203 trip at bed-1 exit > 660 °C, 5 s (same element as TI-203) • TIC-201 alarm LO 405 / HI 435 °C	**Medium 9**

Recommendation

Review the effects of I-2 TSHH-203: the trip stops the sulfur only, and with TV-201 wide open the bed-1 temperature keeps rising after the trip (hot furnace refractory heats the air, see D3). Consider forcing TV-201 closed and cutting the air on I-2.

Response of the action party

Status: ☐ Accepted and done ☐ Accepted with an alternative ☐ Rejected (justification attached) ☐ Open

	Name	Signature	Date
Action party			
HAZOP leader (verification)			
Plant manager (acceptance)			